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Investigating and analyzing the Challenges of Accessing Afghan Cultural Information and Proposing a Solution through the Design and Implementation of a Crowdsourced Platform

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Acknowledgment of Thesis Defense

It is hereby confirmed that the thesis of the honorable **Shah Mahmood** the son of **Mohammad Hashem** titled **Investigating and analyzing the Challenges of Accessing Afghan Cultural Information and Proposing a Solution through the Design and Implementation of a Crowdsourced Platform**. Under the supervision of the respected teacher, Towfiq Amiri, was written and the above-mentioned student has successfully defended his thesis on 2026 / 09 / . The final score is

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With respect,
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Abstract

Afghan cultural information represents an important part of the country's historical identity and collective knowledge; however, access to such information remains challenging because cultural resources are often scattered across different sources, insufficiently documented, and not organized within a unified digital environment. Existing digital platforms provide access to cultural information in some areas, but often offer limited support for community contribution, structured organization, and ongoing improvement of cultural knowledge. This study investigated the challenges affecting access to Afghan cultural information and developed a crowdsourced web-based solution to address the identified challenges. Library research was conducted to examine existing literature, identify relevant challenges, and establish the research gap, while Design Science Research Methodology (DSRM) guided the design and development of the proposed solution. The resulting Afghan Cultural Knowledge Platform provides structured cultural entries with geographical and thematic organization, supporting references, community participation features, user contribution management, and moderation workflows. The platform enables registered users to contribute cultural information through a controlled submission process in which contributions can be reviewed, revised, and managed before publication. Moderators can review submitted content and make appropriate decisions, while administrators manage users, roles, taxonomies, reports, and other platform-level functions. The developed system demonstrates a structured approach to collecting, organizing, reviewing, and presenting Afghan cultural information through community participation. As the platform was not deployed with a large community of users, the study evaluates the developed solution primarily through its implemented features and demonstrated workflows and does not claim measured improvements in cultural preservation, information accessibility, or user participation.

Keywords Cultural Heritage; Crowdsourcing; Cultural Knowledge; Community Participation; Digital Preservation;

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Chapter 1:

Introduction

1.1 Introduction

Cultural heritage is an important part of the identity, history, and collective memory of every society. It includes tangible elements such as historical places, buildings, manuscripts, artworks, photographs, handicrafts, and cultural objects, as well as intangible elements such as traditions, customs, languages, music, poetry, oral stories, ceremonies, local knowledge, and traditional skills. These cultural elements help communities understand their history, maintain their cultural identity, and transfer knowledge and values from one generation to another.

Afghanistan has a rich and diverse cultural heritage formed by different provinces, communities, traditions, languages, historical places, local stories, arts, clothing, food, music, poetry, handicrafts, and social practices. This cultural information is valuable for students, teachers, researchers, writers, cultural institutions, Afghan communities, and other people who want to learn about the country's culture and history.

However, the existence of cultural information does not necessarily mean that it is easy to access. Information about Afghan culture may be found in books, archives, websites, social media, personal collections, and oral sources. Because these sources are separate from one another, users may experience difficulties in finding related information, understanding its cultural or geographical context, identifying its source, or determining its reliability. In addition, some cultural knowledge is held by local people and communities and may remain undocumented or inaccessible to wider audiences.

Digital technologies provide opportunities to collect, organize, preserve, and share cultural information in more accessible ways. Crowdsourcing can also allow people and communities to participate in documenting cultural knowledge instead of depending only on institutions or a small number of content creators. Therefore, this research investigates the challenges of accessing Afghan cultural information, studies how a crowdsourcing approach can help address those challenges, and develops a crowdsourced web-based solution for improving access to Afghan cultural information.

1.2 Problem Statement

Afghan cultural knowledge is distributed across communities, regions, individuals, printed materials, websites, archives, social media, personal collections, and oral sources, which can make it difficult for users to find relevant information, understand its cultural or geographical context, identify its source, and evaluate its reliability. In addition, much cultural knowledge is preserved through personal experience, oral communication, traditions, and local practices and may remain undocumented or inaccessible to students,

researchers, cultural institutions, younger generations, and other interested people. Effective access to Afghan cultural information requires an organized, accessible, and reliable environment in which users can easily discover related cultural knowledge and community members can contribute, correct, and expand existing information while appropriate review processes help maintain its quality. However, Afghan cultural information remains scattered across disconnected and inconsistently organized sources, some materials lack clear context or source information, and community participation in documenting and improving cultural knowledge is still limited. Therefore, there is a need to investigate and analyze these access challenges and develop a centralized crowdsourced web platform that can support the organized documentation, accessibility, participation, and preservation of Afghan cultural information.

1.3 Research Objectives

1.3.1 Main Objective

To investigate and analyze the challenges of accessing Afghan cultural information and develop a crowdsourced web-based solution to address these challenges

1.3.2 Sub Objectives

1. To identify the challenges affecting access to Afghan cultural information.
2. To analyze the identified challenges and their effects on access to Afghan cultural information.
3. To develop a crowdsourced web-based solution that addresses the identified challenges of accessing Afghan cultural information.

1.3.3 Main Research Question

What are the main challenges affecting access to Afghan cultural information, and how can a crowdsourced web-based platform be designed to address these challenges?

1.3.4 Sub Research Questions

1. What are the challenges affecting access to Afghan cultural information?
2. What are the effects of the identified challenges on access to Afghan cultural information?
3. How can a crowdsourced web-based solution be developed to address the identified challenges of accessing Afghan cultural information?

1.4 Research Significance

This research has cultural, social, educational, practical, academic, and technical significance because it focuses on improving the documentation and accessibility of Afghan cultural knowledge. A centralized crowdsourced platform can provide an organized environment for documenting and sharing traditions, oral stories, local practices, historical information, photographs, videos, and other cultural materials from different provinces and communities, while also supporting the preservation and transfer of this knowledge to future generations. It can help students, teachers, researchers, writers, and other people interested in Afghan culture find related information through one system instead of searching across several disconnected sources, with information organized according to factors such as province, category, and topic. The proposed system also connects the identified information-access problem with the design and implementation of a practical web-based solution that allows community members to contribute cultural content, suggest corrections, provide reviews, and report inaccurate or inappropriate materials, while moderation and review processes help improve the reliability and organization of user-submitted information. This participatory approach can give local communities and cultural knowledge holders a more direct role in documenting traditions, crafts, stories, places, languages, food, music, ceremonies, and other forms of cultural knowledge. From an academic and technical perspective, the study also provides a practical example of applying crowdsourcing and web technologies to cultural-information access and may support future research in areas such as digital cultural heritage, community participation, content verification, cultural information organization, and the development of similar cultural platforms.

1.5 Research Methodology

This research uses the library research method to investigate the challenges of accessing Afghan cultural information and the Design Science Research Methodology (DSRM) to guide the design and development of the proposed solution.

1.6 Research Structure

Chapter 1: Introduction

Chapter 2: Literature Review

Chapter 3: Method and Materials

Chapter 4: Results

Chapter 5: Discussion

Chapter 6: Conclusion

References

Terminology

Chapter 2:

Literature Review

2.1 Introduction

Access to cultural information is not only a matter of putting cultural material online. Previous research shows that cultural information may remain difficult to discover when it is dispersed across separate collections, poorly documented, disconnected from its cultural or geographical context, or available only through institutions that do not provide broad public participation. At the same time, digital participation creates its own challenges because information contributed by the public must be organized, reviewed, contextualized, and preserved in ways that maintain its usefulness and reliability. For this reason, the literature reviewed in this chapter is organized around five challenges that are directly related to this research: fragmentation and limited access, insufficient documentation and risk of loss of local cultural knowledge, limited community participation, reliability and verification of crowdsourced information, and lack of cultural, geographical, and source context. (Moustafa, 2016; Constantinidis, 2016; Oomen & Aroyo, 2011; Tan, Yan, & Wang, 2026)

The reviewed studies include research on Afghanistan as well as international work on digital cultural heritage, crowdsourcing, community participation, metadata, provenance, geotagging, digital repositories, and emerging information-access technologies. The purpose of reviewing these studies is not simply to describe existing systems, but to examine how previous researchers understood each challenge, what solutions they proposed or implemented, what results they reported, and what limitations remained. This critical comparison provides the basis for identifying the research gap at the end of the chapter. (Constantinidis, 2016; Daquino, Wigham, Daga, Giagnolini, & Tomasi, 2022; Palombo, 2023; Caridakis, et al., 2024)

2.2 Fragmentation and Limited Access to Afghan Cultural Information

A major challenge in accessing Afghan cultural information is that relevant materials have historically been dispersed, hidden, damaged, privately held, or stored in separate institutions and collections. Moustafa examined the Afghanistan Center at Kabul University (ACKU) and showed how the center developed from efforts to collect Afghan books, documents, newspapers, photographs, and other materials that had been displaced or endangered during decades of conflict. ACKU continued to search for hidden and dispersed Afghan materials, including items held privately, and later digitized large parts of its collection to make previously difficult-to-access resources available to researchers and students. This work demonstrates that digitization and institutional collection can substantially improve access; however, the solution remains centered on an archival and library collection and does not by itself

bring together the wider range of cultural knowledge held across communities, informal sources, and different digital environments. (Moustafa, 2016)

The problem of fragmentation is not limited to Afghanistan. Candela et al. observed that cultural heritage institutions increasingly publish digital collections in different formats, but many of these collections continue to exist as independent data silos. Such separation can hinder collaboration, create duplicated work, produce inconsistencies, and limit the ability to combine information from different sources. Their proposed direction is the use of shared and federated research infrastructures that make it easier to integrate and reuse cultural data. Oomen and Aroyo similarly argued that digitization creates new routes to cultural collections and that open, connected infrastructures can improve accessibility and reuse. These approaches improve institutional interoperability, but they primarily address already digitized collections and do not fully solve the problem of cultural information that remains outside formal repositories or scattered across community and public sources. (Candela, et al., 2025; Oomen & Aroyo, 2011)

More recent work has also explored new ways to search and interact with large cultural collections. Scharnhorst et al. described an engineering approach that combines structured archives, knowledge organization systems, and Retrieval-Augmented Generation to let users ask natural-language questions about specific collections while still seeing the source records used for retrieval. This can make a well-organized collection easier to explore, but it assumes that the underlying material has already been collected, structured, and preserved. Therefore, advanced search technology can improve interaction with existing repositories, but it cannot by itself solve the earlier problem of fragmented or missing cultural information. (Scharnhorst, et al., 2025)

For Afghanistan specifically, Constantinidis proposed a more participatory digital approach in which local people could use mobile devices to record cultural heritage and upload photographs and information to a dedicated website supported by a geographical information system. Such a platform could improve discovery by bringing public contributions into a shared environment and connecting information with place. However, the proposal was primarily a conceptual framework for future Afghan crowdsourcing rather than evidence from a fully implemented national cultural-information platform. Palombo and Yates also show why simple digital aggregation is not enough: in their study of the 'Afghan Genizah,' digitized materials from northern Afghanistan became internationally accessible while remaining physically dispersed, poorly provenanced, and disconnected from their original communities. This suggests that improving access requires not only centralization, but also careful attention to ownership, origin, provenance, and

who actually benefits from digital access. (Constantinidis, 2016; Palombo, 2023)

2.3 Insufficient Documentation and Risk of Loss of Local Cultural Knowledge

A second challenge is that some cultural knowledge is not yet adequately documented in formal archives or digital systems. This is especially important for knowledge carried through oral stories, local traditions, memories, practices, music, folklore, and community experience. Constantinidis argues that digital crowdsourcing can help record both tangible and intangible heritage and proposes the use of ordinary mobile devices to capture cultural material that may otherwise disappear. In the Afghan context, the paper emphasizes that physical destruction, looting, conflict, construction, and other threats can cause permanent cultural loss, making digital recording a useful supporting strategy when physical preservation is difficult. The value of this approach is that documentation does not depend only on professionals being physically present everywhere; however, the quality and completeness of public documentation still depend on participation, organization, and appropriate cultural guidance. (Constantinidis, 2016)

Moustafa's study demonstrates the same preservation problem from the perspective of documentary heritage. The article reports extensive losses to Afghan libraries and museum collections during conflict and shows how Nancy Dupree and later ACKU collected books, newspapers, government documents, photographs, and other materials that might otherwise have disappeared. ACKU's digitization program helped protect and provide access to surviving materials, but the center also faced shortages of funding, trained staff, preservation expertise, and disaster planning. The study therefore demonstrates that collecting and digitizing cultural materials can reduce the risk of loss, while also showing that institutional preservation is difficult to sustain when resources and professional capacity are limited. (Moustafa, 2016)

Research on intangible and endangered cultural knowledge adds another important criticism: documentation alone does not necessarily mean successful preservation. Tan et al., after reviewing 79 studies on digital safeguarding of intangible cultural heritage, argue that technology-focused projects with weak community participation can create a form of 'digital fossilization,' where cultural practices are recorded but separated from the living communities that give them meaning. Shi et al. reached a related conclusion in their review of endangered scripts. They found that 58 percent of reviewed studies focused on foundational digital survival, while only 18 percent reached community- and public-oriented applications. Together, these

studies suggest that preservation should move beyond storing digital representations and should also support continued accessibility, use, and community connection. (Tan et al., 2026; Shi et al., 2026)

Ziku et al. tested crowdsourcing tools for collecting myths, legends, folklore, and local-history stories connected to a heritage site in Rhodes. Their work shows that web-based storytelling and geotagging can provide practical mechanisms for documenting community-related cultural content. However, the authors acknowledge that the stories collected in their case were mainly taken from already documented sources and that the study was less focused on capturing original living-heritage knowledge. This limitation is important for the present research because it shows that providing a contribution form or storytelling tool does not automatically result in the documentation of knowledge that has never been formally recorded. Reaching knowledgeable local contributors and creating appropriate structures for their knowledge remain unresolved parts of the documentation challenge. (Caridakis, et al., 2024)

2.4 Limited Community Participation in Cultural Documentation

Cultural information systems have traditionally been managed by museums, archives, libraries, researchers, and other professional institutions, while the public has often been treated mainly as an audience. Oomen and Aroyo examined the shift toward participatory cultural heritage and identified several ways in which the public can contribute, including correcting or transcribing digitized material, adding contextual stories, complementing collections with new objects, classifying material, and participating in curation. Their analysis shows that crowdsourcing can extend cultural documentation beyond professional staff and allow knowledge from wider communities to enter cultural collections. At the same time, they identify one of the most important practical problems: successful crowdsourcing requires enough knowledgeable and loyal contributors, and participation cannot be assumed simply because a platform is available. (Oomen & Aroyo, 2011)

Seitsonen provides a practical example of what community participation can contribute. In a Finnish conflict-heritage crowdsourcing project, members of the public added nearly two thousand observations and stories to an online map, attached photographs, wrote detailed descriptions, replied to other contributions, and sometimes corrected or challenged existing entries. The contributions also reflected local interpretations of heritage that were not always the same as official institutional views. This demonstrates that communities can provide knowledge and perspectives that formal heritage organizations may not contain. However, participation was strongly

influenced by media attention and follow-up publicity, and much activity occurred in short bursts after public prompts. This indicates that participation is valuable but must be actively encouraged and sustained. (Seitsonen, 2017)

Constantinidis applied this participation problem directly to Afghanistan. The paper proposed that influential 'change-engage agents,' education campaigns, mobile phones, and a dedicated online environment could motivate Afghans to photograph and share cultural heritage that is important to them. The proposal recognizes that the effectiveness of crowdsourcing depends on social motivation and local communication practices rather than technology alone. Its limitation is that the proposed Afghan model was not presented as a completed national platform with measured participation outcomes. Therefore, it provides strong conceptual support for Afghan cultural crowdsourcing, but it leaves open the practical question of how community contribution should be organized in a real, sustainable information system. (Constantinidis, 2016)

Recent studies reinforce the need to design participation around communities rather than treating contributors as simple data providers. Tan et al. argue that communities should participate in the design, implementation, and governance of digital safeguarding, because cultural knowledge remains meaningful only when its bearers retain agency. Ziku et al. also found interest among participants in contributing to local cultural-heritage applications, but their study reported limited prior experience with contribution and emphasized the importance of communication channels, targeted communities, and engagement strategies. These findings show that crowdsourcing can broaden cultural documentation, but participation is uneven and depends on trust, motivation, accessibility, and the perceived value of contribution. (Tan, Yan, & Wang, 2026; Caridakis, et al., 2024)

2.5 Reliability and Verification of Crowdsourced Cultural Information

Opening cultural-information systems to public contribution creates a direct reliability challenge. Oomen and Aroyo identify maintaining a reasonable level of quality as one of the two critical problems in cultural-heritage crowdsourcing. Their review notes that crowdsourced data can contain conflicting information, uncertain annotations, incomplete provenance, and uneven contribution across a collection. They therefore argue for clear reviewing procedures and mechanisms that help communities filter or correct incorrect information. Their examples show that quality control does not have to depend only on institutional experts; technological filters, community agreement, feedback, and contributor reputation can also support quality. However, the paper makes clear that the tension between open participation

and trusted information cannot be removed completely and must be managed as part of the crowdsourcing process. (Oomen & Aroyo, 2011)

Daquino et al. addressed this problem through the design of CLEF, a web-based platform for collaborative cultural-heritage data collection, peer review, and publication. CLEF uses structured templates to improve consistency, supports provenance tracking so changes and sources can be followed, and provides an editorial workflow for reviewing records before and after publication. These mechanisms are relevant because they connect contribution with accountability rather than treating user-generated information as immediately trustworthy. Nevertheless, CLEF also has limitations: its broader end-user usability still required testing, some parts of the system depend on external services and ontologies, and migration from existing museum systems was not yet implemented. The study therefore demonstrates a strong technical approach to reliability, but not a universal solution for all community-based cultural contexts. (Daquino, Wigham, Daga, Giagnolini, & Tomasi, 2022)

Ziku et al. also describe practical moderation mechanisms in crowdsourced storytelling applications. In their case study, contributions were reviewed for quality and relevance, account verification was required for uploading, and the authors suggested that larger systems could combine teams of moderators, user reporting, submission guidelines, automated checks, and expert review. They also note that cultural-heritage crowdsourcing continues to face credibility, liability, and validation problems. This is particularly important for living heritage and stories, where information may be subjective, collectively owned, or difficult to verify through a single formal source. The study therefore supports moderation and review but also shows that verification must be flexible enough to handle different kinds of cultural knowledge. (Caridakis, et al., 2024)

The Finland case shows the same reliability problem from another angle. Seitsonen found that contributors sometimes corrected or questioned other users' entries, which demonstrates the value of community review, but geographical accuracy was inconsistent: in one analysed sample, only about ten percent of user-marked points were within 500 metres of the actual locations. Some inaccuracy may have been intentional to protect sensitive sites, showing that a contribution can be inaccurate for reasons other than lack of knowledge. Palombo and Yates further show that reliability also depends on provenance. Their analysis of unprovenanced Afghan manuscripts demonstrates that digital presentation can give material an appearance of coherence and legitimacy even when its origin and movement remain uncertain. Reliable cultural information therefore requires both content review and transparent source/provenance information. (Seitsonen, 2017; Palombo, 2023)

2.6 Lack of Cultural, Geographical, and Source Context

Cultural information loses part of its meaning when it is separated from the place, community, source, or relationships that give it context. Constantinidis emphasizes the geographical dimension of heritage by proposing that crowdsourced Afghan photographs and information be stored in a GIS-supported environment. Geotagged records could allow users to view heritage on maps, connect text and images with location, and search cultural material in relation to its spatial setting. The paper also notes, however, that free-style public tagging can create management and retrieval difficulties. This suggests that geographical access is useful, but it should be combined with organized categories, controlled information structures, and contextual description rather than relying only on unrestricted tags. (Constantinidis, 2016)

Ziku et al. similarly examined the role of geotagging, storytelling, and metadata in the digital documentation of local history and living heritage. Their study found that existing web tools can connect stories with location, but it also identifies a deeper problem: living heritage contains complex concepts such as community agency, shared authorship, and collective practice that are difficult to represent with ordinary metadata fields. The authors note that documentation structures for participatory living heritage remain insufficiently standardized and that common metadata elements such as creator, date, rights, and coverage may be difficult to apply. Therefore, contextual organization is not simply a matter of adding more fields; the information model must reflect the characteristics of cultural knowledge itself. (Caridakis, et al., 2024)

The importance of source context is especially clear in Palombo and Yates' study of the 'Afghan Genizah.' They argue that unprovenanced cultural objects are decontextualized objects, and that digitization can sometimes conceal rather than solve the fact that a collection is physically dispersed or removed from its original environment. Their analysis also questions whether international digital access actually benefits people and researchers in Afghanistan when the materials, infrastructure, and control are located elsewhere. This study is important for the present research because it shows that a culturally responsible information platform should make origins and sources visible instead of presenting cultural information as detached digital content. (Palombo, 2023)

Other digital-heritage research supports the same concern from the perspective of representation. Wang et al., in their review of immersive technologies, found that digital tools can improve education, dissemination, and engagement but can also create risks of cultural misinterpretation, loss of authenticity, and superficial understanding when digital representations are separated from the

meaning of the original heritage. Tan et al. likewise warn that cultural material can be decontextualized, commodified, or misrepresented when community knowledge is separated from its bearers. Shi et al. argue that digital preservation should retain cultural and historical context rather than treating heritage as purely technical data. These studies show that accessibility must be balanced with contextual accuracy and cultural meaning. (Du, Li, Li, Wang, & Zhang, 2025)

2.7 Research Gap

The reviewed literature demonstrates significant progress in different parts of the cultural-information access problem. Digital libraries and research infrastructures have improved access to previously hidden or separated collections; crowdsourcing has enabled the public to add stories, annotations, corrections, photographs, and contextual knowledge; geotagging and metadata have improved geographical discovery; and systems such as CLEF have shown how provenance and peer review can support more reliable collaborative data. However, these solutions are usually developed for particular institutions, collections, technologies, or case studies, and the literature continues to identify unresolved problems involving fragmentation, sustained participation, quality control, provenance, cultural context, and the documentation of living or previously undocumented knowledge. (Moustafa, 2016; Candela, et al., 2025; Oomen & Aroyo, 2011; Daquino, Wigham, Daga, Giagnolini, & Tomasi, 2022; Caridakis, et al., 2024)

The Afghanistan-focused studies make this gap more specific. Moustafa documents substantial efforts to rescue, collect, digitize, and provide access to Afghan documentary heritage through ACKU, but this work is primarily an institutional library and archive model. Constantinidis proposes a crowdsourced digital strategy for Afghan cultural heritage in which local people contribute photographs and geographically referenced information, but the study remains mainly a proposed framework and leaves questions of long-term participation, information management, retrieval, ethics, and contributor knowledge for further research. Palombo and Yates show that putting Afghan heritage online can still leave problems of provenance, unequal access, dispersal, and separation from original communities unresolved. Therefore, the reviewed Afghan literature shows important preservation and digitization initiatives and a strong conceptual argument for crowdsourcing, but it does not demonstrate an implemented general-purpose platform that brings these different requirements together for broad Afghan cultural information. (Moustafa, 2016; Constantinidis, 2016; Palombo, 2023)

Within the literature reviewed in this chapter, the remaining gap is therefore not simply the absence of digital cultural material or the absence of

crowdsourcing. The unresolved issue is the lack of an integrated Afghan-focused approach that combines organized access to cultural information with community contribution, documentation of local knowledge, cultural and geographical context, source transparency, and mechanisms for reviewing and improving contributed information. International studies provide useful individual solutions to these problems, but they are commonly tested in other countries, within specific institutional collections, or for narrow types of cultural content. The present research addresses this gap by designing and implementing a centralized crowdsourced web platform intended to bring these requirements together for Afghan cultural information, while focusing specifically on improving access, documentation, participation, organization, and reliability rather than attempting to solve every broader problem of cultural-heritage preservation. (Oomen & Aroyo, 2011; Daquino, Wigham, Daga, Giagnolini, & Tomasi, 2022; Seitsonen, 2017; Caridakis, et al., 2024; Constantinidis, 2016)

Chapter 3:

Method and Materials

3.1 Introduction

This chapter explains how the research was carried out and what materials were used to develop the proposed Afghan Cultural Information Crowdsourcing Platform. It first presents the research design and the research methods used to investigate the challenges of accessing Afghan cultural information. It then explains how the collected documentary evidence was analyzed and how the findings were translated into system requirements. After that, the chapter presents the system development methodology, the functional and non-functional requirements, the system architecture, and the main programming languages, frameworks, libraries, services, and tools used in the project. Finally, it describes the database design, the implementation of the backend and frontend, and the security measures applied to the system.

3.2 Research Design

This study follows an applied research and system development design. The purpose of the research is not only to describe the difficulties surrounding access to Afghan cultural information, but also to use the findings of that investigation to produce a practical software artifact that responds to the identified problems. The study therefore has two connected parts. The first part investigates the existing challenges through published academic literature, with particular attention to Afghanistan and to related international work on digital cultural heritage and crowdsourcing. The second part uses the findings of the investigation to design and develop a Persian-language crowdsourced web platform for Afghan cultural information.

Because the practical contribution of the study is a software artifact, the development-oriented part of the research is organized using Design Science Research. Design Science Research is appropriate in information-systems research when knowledge is produced through the design, construction, demonstration, and evaluation of an artifact intended to address a recognized problem (Peppers, Tuunanen, Rothenberger, & Chatterjee, 2007). In this study, the literature establishes the problem and its main dimensions, while the proposed platform represents the artifact developed in response to those findings. This creates a direct connection between the research problem, the literature review, the system requirements, and the final implementation.

3.3 Research Methods

Two complementary methods are used in this study: the library research method and the Design Science Research Method. The library research method provides the theoretical and documentary evidence needed to understand the problem, identify recurring challenges, examine previous approaches, and establish the research gap. The Design Science Research

Method provides the process through which those findings are transformed into the requirements, design, implementation, demonstration, and evaluation of the proposed software artifact. The two methods therefore serve different but connected purposes: one investigates the problem from existing knowledge, while the other guides the creation of the solution.

3.4 Library Research Method

The library research method involved reviewing academic journal articles, conference papers, and scholarly studies related to Afghan cultural heritage, digital cultural heritage, crowdsourcing, community participation, cultural documentation, digital preservation, provenance, information reliability, and cultural-information access. Priority was given to studies directly related to Afghanistan, while international studies were included when they addressed problems or solutions relevant to the research topic.

The literature review served four main purposes. First, it identified the principal challenges affecting access to Afghan cultural information. Second, it examined how previous researchers and cultural-heritage projects had addressed similar problems through digitization, crowdsourcing, community participation, structured metadata, provenance, moderation, and other digital approaches. Third, it critically compared the strengths and limitations of these approaches rather than presenting the reviewed studies as separate summaries. Finally, it established the research gap that justified the proposed platform and provided evidence for the requirements later defined for the system.

3.5 Design Science Research Method

The Design Science Research Method was used to transform the knowledge obtained through the library research method into a concrete web-based artifact. Design Science Research focuses on solving practical problems through the creation and study of artifacts such as methods, models, processes, or software systems. It is suitable for this research because the study does not stop after identifying the challenges of accessing Afghan cultural information; it also designs and implements a working crowdsourced platform intended to address important parts of those challenges (Peffer, Tuunanen, Rothenberger, & Chatterjee, 2007).

The study follows the six activities described in the Design Science Research Methodology proposed by Peffer et al. (2007). In the first activity, problem identification and motivation, the literature was used to establish the difficulties surrounding fragmented access, insufficient documentation, limited community participation, reliability and verification, and loss of cultural, geographical, and source context. In the second activity, definition of the objectives of a solution, these findings were translated into the research

objectives and into the functional and non-functional requirements of the platform. In the third activity, design and development, the product requirements, architecture, database, user interface, backend, frontend, content workflow, and security mechanisms were designed and implemented. In the fourth activity, demonstration, the completed functions of the platform are demonstrated through the main user, moderator, and public workflows presented in Chapter 4. In the fifth activity, evaluation, the implemented behavior is examined against the research objectives, identified challenges, and system requirements; this study does not depend on a questionnaire-based user evaluation. In the sixth activity, communication, the research problem, method, artifact, results, and discussion are documented in this monograph.

The application of Design Science Research in this study is iterative rather than strictly linear. Decisions concerning the data model, authentication, moderation, geographical classification, content versioning, internal links, and other system functions were refined as development progressed.

3.6 Data Collection and Analysis

The data used in the investigative part of the study were documentary and mainly qualitative. The first source was the academic literature reviewed in Chapter 2, including Afghanistan-focused studies and international research on digital cultural heritage and crowdsourcing. The second source was the set of systems, platforms, and technical approaches described in those studies, which provided evidence about what previous solutions had attempted, what benefits they reported, and which limitations remained. The third source was the official technical documentation of the technologies selected for the project, which was consulted when technical feasibility, framework behavior, security practices, and implementation decisions had to be defined.

Because the collected academic material was qualitative, thematic analysis was used to identify and organize recurring patterns in the reviewed evidence. Thematic analysis provides a systematic way to identify, analyze, and report patterns or themes within qualitative material (Braun & Clarke, 2021). The reviewed studies were examined for recurring access problems, limitations, proposed solutions, and unresolved issues. Related findings were then grouped into five themes: fragmentation and limited access to Afghan cultural information; insufficient documentation and risk of loss of local cultural knowledge; limited community participation in cultural documentation; reliability and verification of crowdsourced cultural information; and lack of cultural, geographical, and source context.

The themes were then translated into system requirements. Fragmentation and limited access supported the requirement for a centralized, searchable, and consistently organized platform. Insufficient documentation and limited

participation supported user contribution, structured Cultural Entries, media, sources, and community-based correction mechanisms. Reliability concerns supported moderation before publication, role-based review, content versioning, reports, source information, self-approval prevention, and audit records. The need for cultural and geographical context supported structured categories, content types, tags, managed provinces and districts, geographic scope, location details, source types, and manual internal links between related Cultural Entries. These requirements were recorded and refined through the Product Requirements Document, architecture specifications, database design, technology specification, and design system produced during development.

3.7 System Development Methodology

The system was developed using an iterative and incremental development approach. In this approach, software is divided into smaller parts that can be designed, implemented, checked, and refined in successive cycles instead of attempting to complete the entire application in one fixed sequence. This approach was suitable for the project because the overall research problem and first-version scope were known early, while detailed decisions became clearer as the literature, database relationships, content workflow, security requirements, and user-interface behavior were developed.

The development began with the approved database design and the initial Prisma migration, followed by authentication and user-role protection. The shared taxonomy for provinces, districts, categories, content types, and tags was then established before Cultural Entry drafts and media handling were added. The next stages covered submission and moderation, public content pages, Persian search and filtering, bookmarks, ratings and reviews, correction suggestions and reports, administration and audit logs, and finally testing and deployment preparation. Within these stages, individual features were revised when later decisions required changes, such as adding Google and Facebook authentication, mandatory email verification for contribution-related actions, the simplified geographical-scope model, private bookmarks, Persian public slugs with stable internal keys, and Wikipedia-style internal links between published Cultural Entries.

Several development documents were maintained alongside the code so that the system remained consistent as it evolved. The Product Requirements Document defined the product scope, user roles, workflows, functional requirements, non-functional requirements, and business rules. The architecture documents defined module boundaries, request flow, authentication, authorization, API conventions, and implementation order. The database design described the entities, relationships, constraints, indexes, and preservation rules. The technology specification documented the selected

stack and external services, while the design system defined the visual principles and Persian right-to-left interface. These documents were updated during development rather than treated as fixed descriptions that could not change.

3.8 System Requirements

The system requirements describe what the proposed platform must do and the quality characteristics it must maintain. They were derived from the challenges identified in the literature, the research gap, and the practical scope of the first version. Following standard software-engineering practice, the requirements are divided into functional requirements, which describe the behavior and services of the system, and non-functional requirements, which describe qualities such as usability, security, reliability, performance, accessibility, and maintainability (Sommerville, 2015)

3.9 Functional Requirements

The first group of functional requirements concerns user access and account management. Visitors must be able to register with email and password and to authenticate through the supported Google and Facebook social-login providers. Email/password accounts must support email verification, login, logout, password reset, and password change, while the platform must issue its own access and refresh tokens after successful authentication. The system must distinguish the roles USER, MODERATOR, and ADMIN, block suspended accounts from protected operations, and enforce role and ownership rules on the server. Contribution-related actions require a verified email address, while public reading functions remain accessible without authentication.

The second group concerns cultural-content creation. Every registered user must be able to create a Cultural Entry, save it as a draft, edit a personal draft, preview it, and submit it for moderation. A Cultural Entry contains a Persian title, short summary, structured rich-text content, content type, category, and geographical classification, together with optional tags, district and local-place information where appropriate, sources, images, and one supported YouTube video. The geographical model must allow an entry to belong to one specific province, to represent Afghanistan-wide content, or to have no meaningful geographical dependency. Province selection is therefore required only for province-scoped entries. The system must also support manually created internal links from selected text in one Cultural Entry to another published Cultural Entry, prevent self-links, and preserve stable internal references independently from the public URL slug.

The third group concerns moderation and preservation of cultural information. Submitted content must not become public until it has been reviewed. A moderator must be able to approve a submission, request changes, or reject it, and a reason is required for rejection or requested changes. A moderator cannot approve their own contribution. Approval immediately publishes the content, while requested changes allow the contributor to revise and resubmit it. Important submitted and published states must be preserved through ContentVersion records so that accepted corrections and administrator edits do not silently overwrite historical information. Published content may later be hidden, restored, or archived according to moderation decisions.

The fourth group concerns public discovery and community interaction. Visitors must be able to read published Cultural Entries and search normalized Persian content by keyword. Public browsing and filtering must support category, content type, tags, geographical scope, and province for province-scoped entries. Registered users must be able to rate published content on a 1-to-5 helpfulness scale, write public reviews, privately bookmark published entries, suggest corrections, and submit private reports about inaccurate, inappropriate, misleading, culturally sensitive, or otherwise problematic content. Corrections must be reviewed before changing published information, and accepted corrections must create a new content version. Reports must be investigated by moderators rather than automatically removing the reported content.

The final group concerns administration, taxonomy, media, and traceability. Administrators must be able to manage users, roles, account suspension, provinces, districts, categories, content types, tags, and important platform records. Image uploads must be validated before being sent to Cloudinary, with the first version allowing up to six images per Cultural Entry and a maximum size of 4 MB per image. Video files are not uploaded directly; the backend validates supported YouTube links and stores only the information required for embedding the video. Important moderation and administrative actions must be recorded in audit logs so that decisions such as approval, rejection, requested changes, hiding, restoration, accepted corrections, report resolution, role changes, and suspension can be traced.

3.10 Non-Functional Requirements

The non-functional requirements define the quality expected from the platform. Usability requires a simple interface that does not demand advanced technical knowledge from contributors or readers. The first version must use Persian throughout the interface, use right-to-left layout, handle Persian and Arabic character variations correctly, and display readable content on different screen sizes. Accessibility requires clear labels, keyboard-accessible controls,

meaningful alternative text for images, and sufficient visual contrast. Performance requires public pages, filtering, and common dashboard operations to respond efficiently, while images are delivered in optimized forms such as thumbnails when appropriate.

Security requires protected authentication, hashed passwords and tokens, server-side authorization, validation, controlled file uploads, rate limiting, safe error responses, and secure management of secrets. Privacy requires private account information and reports to remain inaccessible to unauthorized users. Reliability and data integrity require valid workflow transitions, transactions for related multi-record changes, consistent ratings and version history, and preservation of moderation and audit information. Maintainability requires a modular codebase, clear separation of frontend, backend, and database responsibilities, consistent TypeScript conventions, and reusable validation. Search-engine visibility is also important because published cultural information is intended to be publicly discoverable; therefore, public entries use readable Persian slugs and server-rendered pages where appropriate.

3.11 System Architecture

The system follows a client-server architecture implemented as two separate applications inside one pnpm workspace repository. The frontend is built with Next.js and is responsible for presentation and user interaction. The backend is a modular NestJS monolith that exposes a versioned REST API and is responsible for authentication, authorization, validation, business rules, moderation, corrections, reports, ratings, reviews, bookmarks, media handling, administration, and database access. PostgreSQL is the application database and is accessed through Prisma. Cloudinary is used for image storage, SMTP is used for transactional email, and YouTube content is included through validated links rather than direct video uploads.

The frontend and backend are deliberately kept separate. Public cultural pages, the Persian right-to-left interface, authentication pages, contribution forms, Tiptap editing, search and filter controls, and the user, moderator, and administrator dashboards are implemented in Next.js. Public reading pages use Next.js Server Components where suitable, while interactive features use Client Components. Interactive server data is managed through TanStack Query, forms through React Hook Form and Zod, and temporary shared interface state through Zustand. The frontend does not access Prisma or PostgreSQL directly.

The NestJS backend follows a modular structure in which controllers receive HTTP requests, services apply business rules and perform database operations, DTOs define and validate accepted request data, and guards or policies enforce authentication, role, ownership, verified-email, and workflow

rules. Authentication is global by default: protected API routes require a valid access token unless a route is explicitly public. This ensures that hiding a button or route in the frontend is treated only as a user-interface decision and never as a replacement for backend authorization.

The flow of a normal request is therefore straightforward. A user performs an action in the Next.js frontend, which sends a request to the NestJS REST API. The backend authenticates and authorizes the request where required, validates the supplied data, and passes the request from the controller to the relevant service. The service applies the business rules and uses Prisma to read or write PostgreSQL data, sometimes within a transaction when several related records must change together. The backend then returns a structured response, and the frontend updates the page or dashboard. This separation keeps presentation, business rules, and persistence independent and makes the application easier to maintain and extend.

3.12 System Use Case Diagram

The use case diagram presents the main actors and interactions within the platform. The system contains four main actors: Visitor, Registered User, Moderator, and Administrator. Visitors can explore published cultural information, while registered users can contribute content and participate in community activities. Moderators are responsible for reviewing submitted content and managing improvement processes, while administrators manage platform-level operations.

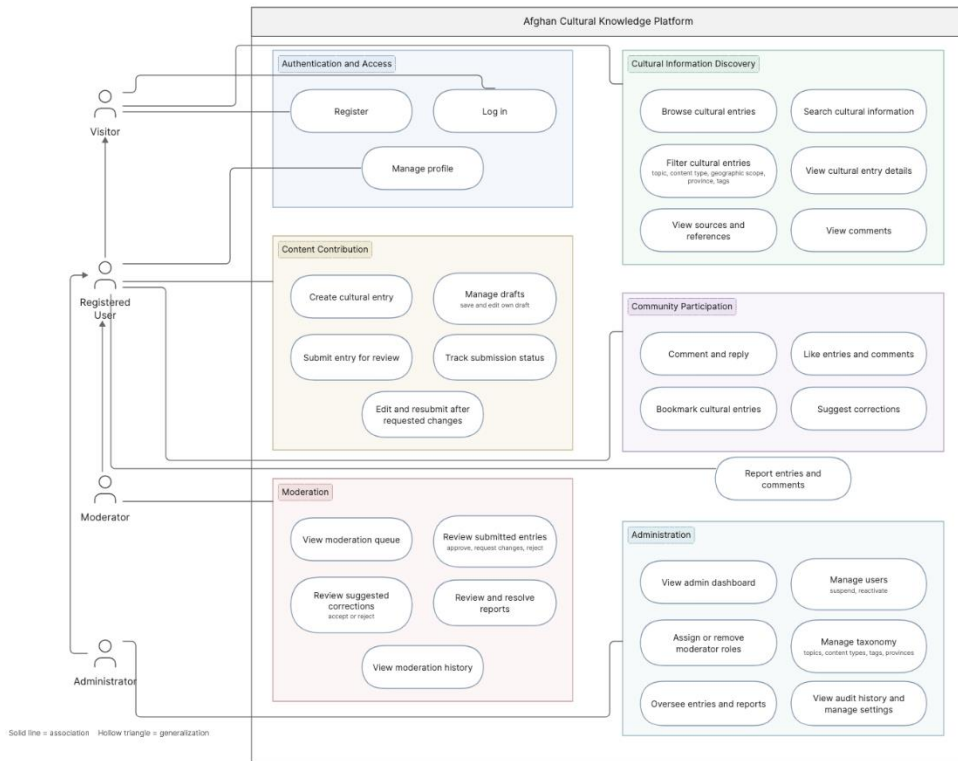


Figure 3.12.1 Entity-Relationship Diagram of the Afghan Cultural Information Crowdsourcing Platform

3.13 Materials, Languages, Frameworks, and Tools

The platform was developed with a set of technologies selected to support a Persian right-to-left public website, community contribution, moderation workflows, structured cultural content, secure authentication, relational data, and maintainable development by a single developer. The following subsections introduce the principal technologies, explain what each one is, and describe how it is used in the project.

3.13.1 TypeScript

TypeScript is a statically typed programming language that extends JavaScript by adding a type system and compile-time checking while remaining compatible with the JavaScript ecosystem. In this project, TypeScript is the main programming language for both the frontend and backend. It is used in

Next.js pages and components, frontend validation and API types, NestJS modules, controllers, services, DTOs, guards, and other backend code. Using TypeScript across both applications reduces differences between the client and server development environments and helps detect invalid data shapes and many programming errors before runtime (TypeScript Documentation, n.d.).

3.13.2 Next.js

Next.js is a React framework for building web applications with file-based routing, server and client rendering, layouts, metadata support, and other web-development features. The project uses the Next.js App Router for the complete frontend. It is used for the public cultural-information website, Cultural Entry detail pages, search and filtering pages, authentication screens, contribution forms, and the user, moderator, and administrator dashboards. Server Components are preferred for public reading pages because they support efficient rendering and public discoverability, while Client Components are used only where browser interaction is required, such as forms, dialogs, ratings, and the rich-text editor (Next.js Docs, n.d.)

3.13.3 Tailwind CSS

Tailwind CSS is a utility-first CSS framework that provides small reusable classes for styling directly in application markup. In the project it is used to implement responsive layouts, spacing, typography, colors, forms, dashboards, content cards, and right-to-left presentation. The project design system defines the visual rules, including a restrained cultural color palette, the Estedad font family, an eight-point spacing system, readable article widths, responsive grids, and subtle interface decoration. Tailwind CSS provides the styling mechanism through which these design rules are applied consistently across the frontend (Tailwind CSS, n.d.)

3.13.4 shadcn/ui

shadcn/ui is a collection of customizable React user-interface components whose source code becomes part of the application rather than being used only as a closed external component package. In this project it provides reusable building blocks such as buttons, inputs, text areas, forms, dialogs, dropdown menus, tables, cards, badges, sidebars, pagination controls, skeletons, and chart-related components. These components are adapted to the Persian right-to-left design system so that public pages and dashboards remain visually consistent while still allowing project-specific customization (shadcn/ui, n.d.)

3.13.5 React Hook Form and Zod

React Hook Form is a React library for managing form values, field state, submission state, and validation errors with limited unnecessary re-rendering.

It is used for registration, login, profile editing, Cultural Entry creation, source information, correction suggestions, reports, and administrative forms. Zod is a TypeScript-oriented schema-validation library used to define the rules that frontend values must satisfy before a request is sent. In the project, React Hook Form manages the behavior of the form while Zod defines client-side validation schemas, providing immediate Persian validation feedback to users. Backend validation still remains authoritative and cannot be replaced by frontend validation (React Hook Form Docs, n.d.; Zod, n.d.) .

3.13.6 TanStack Query and Zustand

TanStack Query is a server-state management library for React applications. It manages data received from the NestJS API in interactive frontend features and provides caching, request deduplication, loading and error states, mutations, invalidation, and background refresh. In this project it is used for interactive data such as entries, moderation queues, corrections, reports, ratings, reviews, users, and other dashboard information. Zustand is a lightweight state-management library used only for temporary shared client-interface state, such as interface controls or multi-step form progress when several components need the same local state. API collections are not duplicated in Zustand because those remain under TanStack Query (TanStack React Query Docs, n.d.; Zustand, n.d.).

3.13.7 Tiptap

Tiptap is an extensible rich-text editor framework based on structured document content. It is used for writing Cultural Entries such as articles, oral histories, traditions, practices, and place descriptions. The editor supports paragraphs, headings, emphasis, lists, block quotations, links, alignment, right-to-left direction, undo and redo, and other controlled formatting needed for Persian cultural content. The editor content is stored as structured Tiptap JSON in PostgreSQL, while a plain-text representation is also generated for search, previews, word counts, moderation, and content comparison. The project also extends Tiptap with manual Wikipedia-style internal links that allow selected text to reference another published Cultural Entry without depending only on an ordinary external URL (Tiptap Docs, n.d.).

3.13.8 NestJS

NestJS is a server-side Node.js framework for building structured applications with TypeScript. It organizes backend code through modules, controllers, providers or services, dependency injection, guards, pipes, filters, and interceptors. In this project, NestJS implements the complete REST API and contains the main business rules. Separate modules handle authentication, users, Cultural Entries, bookmarks, moderation, corrections, reports, ratings,

reviews, taxonomy, media, auditing, and administration. Controllers receive requests, DTOs and validation pipes validate them, services apply business rules and use Prisma, while guards and policies enforce role, ownership, verified-email, and workflow requirements. This modular structure keeps the backend organized without the additional operational complexity of microservices (NestJS Docs, n.d.).

3.13.9 PostgreSQL

PostgreSQL is an open-source relational database management system that stores information in related tables and supports constraints, indexes, transactions, JSON data types, and advanced querying. It is used as the only application database in this project. PostgreSQL stores users, authentication records, Cultural Entries, versions, moderation reviews, provinces, districts, categories, content types, tags, media metadata, sources, ratings, reviews, bookmarks, corrections, reports, audit logs, and other related data. It was selected because the platform contains many relationships and workflow rules that benefit from referential integrity, unique constraints, indexes, and transactions. PostgreSQL is also used for the first version of Persian search over normalized text (PostgreSQL Docs, n.d.).

3.13.10 Prisma ORM

Prisma is an Object-Relational Mapping tool and database toolkit designed for TypeScript and JavaScript applications. It provides a schema language for defining models and relations, generates a typed database client, and manages database migrations. In this project, Prisma connects the NestJS backend to PostgreSQL. Services use Prisma to query and update the database, while migrations preserve changes to the database structure. Important workflows such as publication, accepted corrections, report resolution, version creation, and rating updates can use Prisma transactions so that related database changes succeed or fail together (Prisma, n.d.) .

3.13.11 REST API and JSON

A REST API is an HTTP-based communication style in which clients interact with server resources through endpoints and standard methods such as GET, POST, PATCH, and DELETE. JSON is a lightweight text format commonly used to exchange structured data between applications. In this project, the Next.js frontend communicates with the NestJS backend through a versioned REST API under the /api/v1 path, and request and response bodies are normally represented as JSON. This separation allows the frontend and backend to be developed independently while keeping final validation, authorization, and business rules in the backend.

3.13.12 Passport, JWT, and OAuth

Passport is an authentication middleware system that provides reusable authentication strategies, while JSON Web Token (JWT) is a compact signed token format used to represent authenticated identity and related claims. The backend uses Passport with JWT to authenticate protected API requests. After successful authentication, the system issues a short-lived access token and a longer-lived refresh token. The current design uses an access-token lifetime of approximately fifteen minutes and a refresh-token lifetime of seven days. The access token is sent with protected API requests, while the refresh token is stored in a secure HTTP-only cookie and its database representation is stored as a hash.

The system also supports Google and Facebook OAuth authentication. OAuth provider accounts are linked to the local User record through OAuthAccount records, and verified provider emails may be used for safe account linking according to the application rules. Provider tokens are not used as the platform session. After successful Google or Facebook authentication, the NestJS backend still issues the platform's own access and refresh tokens, so authorization and session behavior remain consistent regardless of the login method.

3.13.13 Argon2id

Argon2id is a memory-hard password-hashing function designed to make password cracking more expensive by requiring significant memory as well as computation. It combines properties of the Argon2i and Argon2d variants and is recommended as a general-purpose Argon2 mode in RFC 9106. In this project, Argon2id is used to hash local account passwords before storage. Plain-text passwords are never stored in PostgreSQL, and OAuth-only users may not have a local password hash at all.

3.13.14 Multer and Cloudinary

Multer is middleware used by the NestJS file-upload process to receive multipart/form-data uploads, while Cloudinary is an external media-management service used to store and deliver images. In this project, the frontend first shows a local preview and then sends the image to NestJS. The backend validates the file type and size before uploading it to Cloudinary, and the returned secure URL, public identifier, dimensions, caption, alternative text, permission information, display order, and thumbnail information are stored in PostgreSQL. The first version allows up to six images per Cultural Entry, with a maximum size of 5 MB per image (NestJS Docs, n.d.; Cloudinary, n.d.)

Video is handled differently. The platform does not upload or store video files. A contributor may attach a supported YouTube link, which the backend validates before extracting and storing the video identifier and related metadata. The public frontend then displays the video through an embedded player. Restricting the first version to validated YouTube links avoids the storage, processing, and bandwidth requirements of operating a separate video-hosting service.

3.13.15 Nodemailer and SMTP

Nodemailer is a Node.js library for sending email, and SMTP is the standard protocol used by mail clients and applications to submit outgoing email through a configured mail server. In this project, NestJS uses Nodemailer with SMTP for transactional authentication messages. These messages include password-reset email and the email-verification flow required for local email/password accounts. The SMTP provider is configured through environment variables so that mail-server credentials and provider details can change without modifying the application code (Nodemailer, n.d.).

3.13.16 Swagger and OpenAPI

OpenAPI is a standard format for describing HTTP APIs, while Swagger is a common set of tools used to generate and interact with OpenAPI documentation. The NestJS backend uses its Swagger integration to document API routes, accepted request data, authentication requirements, and response structures. During development, the generated API documentation provides a convenient way to inspect and test the REST interface and helps keep the contract between the frontend and backend understandable (NestJS Docs, n.d.).

3.13.17 Testing Technologies

Jest, Supertest, and Playwright are used for different levels of testing. Jest is a JavaScript and TypeScript testing framework used for unit tests of isolated backend logic such as normalization, slug generation, workflow rules, and permission checks. Supertest is used with Jest to send requests to the NestJS application and verify API behavior in integration tests. Playwright is a browser-automation framework used for end-to-end testing of important user workflows, including registration and login, Cultural Entry creation, submission, requested changes, resubmission, moderation approval, public discovery, and correction or report submission. Using these tools at different levels helps test both isolated logic and complete workflows (NestJS Docs, n.d.; Playwright, n.d.).

3.13.18 Development Environment and Project Tools

The project is organized as a pnpm workspace monorepo containing separate Next.js and NestJS applications together with a small shared contracts package. pnpm manages project packages and the lockfile so that dependency installation remains reproducible across the repository. Biome is configured at the repository root as the formatter and linter for both applications, helping maintain consistent source-code style and basic static checks. Git and GitHub are used for version control and preservation of the project history, while environment files provide local configuration for PostgreSQL, JWT secrets, OAuth credentials, Cloudinary, SMTP, and other services. Development and verification also make use of Prisma migration commands, Prisma Studio, Swagger, and standard build, type-check, test, and lint commands.

3.13.19 Database Design

The database was designed in PostgreSQL and implemented through Prisma according to the requirements and workflows of the platform. All major records use UUID primary keys, timestamps are stored in UTC, Prisma models use singular PascalCase naming, and database tables and columns are mapped to snake_case. Public Cultural Entries also use two additional identifiers with different purposes: a stable lowercase Latin key used by seed and import tooling, and a unique normalized Persian slug used as the canonical public URL. The key remains stable for internal tooling, while the public slug gives readers a meaningful Persian address.

Version one contains twenty-four schema entities: User, OAuthAccount, CulturalEntry, ContentVersion, ModerationReview, Province, District, Category, ContentType, Tag, EntryTag, Image, YouTubeVideo, Source, Rating, PublicReview, Bookmark, CorrectionSuggestion, Report, RefreshSession, EmailVerificationToken, PasswordResetToken, AuditLog, and EntryReference. CulturalEntry is the central content entity. A user can author entries, while entries are classified by category, content type, geographic scope, optional province and district, and tags. Images, an optional YouTube video, sources, ratings, public reviews, bookmarks, corrections, reports, moderation records, versions, and internal entry references are connected to the Cultural Entry through relational links.

The geographical model deliberately avoids forcing every piece of cultural information into one province. The GeographicScope value may be PROVINCE, NATIONAL, or NONE. Province-scoped content requires a province and may optionally use a managed district and free-text local location. National content represents cultural information relevant across

Afghanistan, while NONE is used when geographical classification would not be meaningful. This design keeps geographical filtering useful without inventing a location for content that is national or non-geographical.

Content history and reliability are supported through several related entities. ContentVersion stores structured snapshots of important submitted and published states. ModerationReview records review decisions and comments. CorrectionSuggestion represents proposed changes that must be reviewed before publication. Report stores private complaints and their resolution. AuditLog records important moderation and administrative actions. EntryReference stores manual internal links from one Cultural Entry to another, allowing the system to preserve Wikipedia-style relationships independently from the visible anchor text or public slug.

Database constraints and indexes protect the quality and performance of the data. User email addresses, Cultural Entry keys and slugs, OAuth provider identities, bookmarks, and several other combinations use uniqueness rules where duplicates would be invalid. Ratings use a 1-to-5 scale, and rating aggregate information is updated together with rating changes. The database design preserves moderation and published history, uses soft deletion or status changes where historical records must remain, and does not support physical user deletion in version one. Indexes support public listing, Persian search, publication ordering, province and geographic-scope filters, categories, content types, tags, author dashboards, moderation queues, reports, corrections, ratings, reviews, bookmarks, OAuth accounts, internal entry references, email verification tokens, and audit logs.

3.13.19.1 Entity-Relationship Diagram

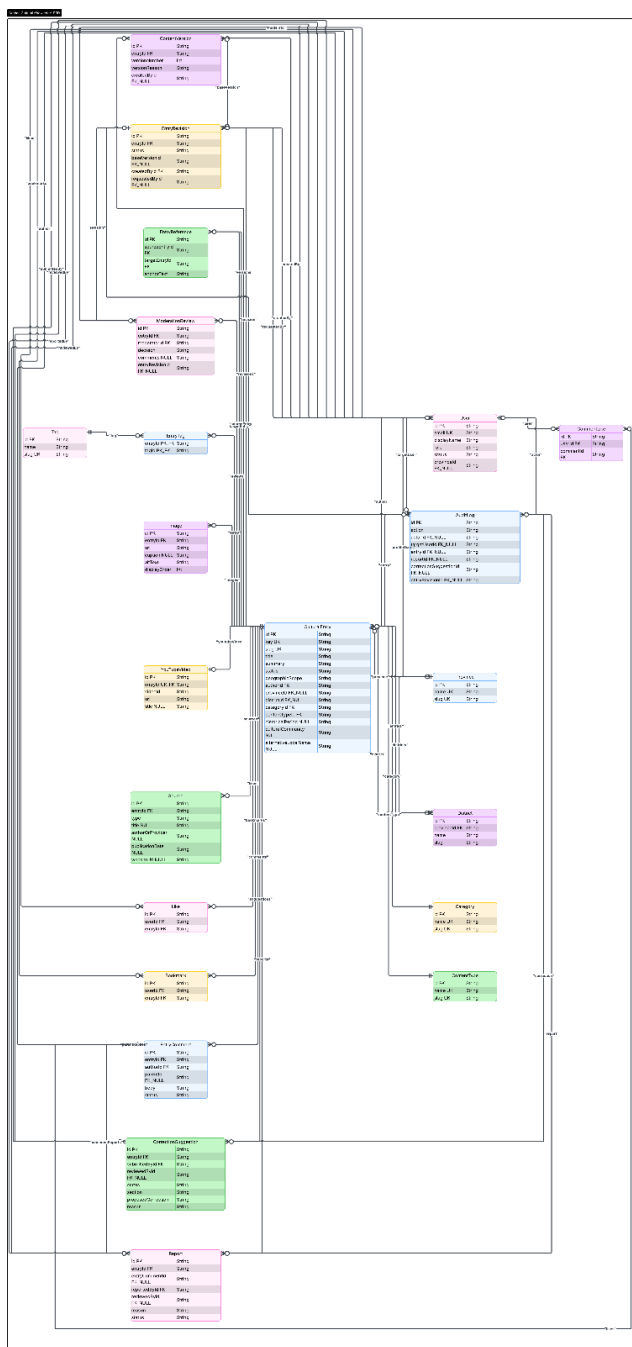


Figure 3.13.1.19.1 Entity-Relationship Diagram of the Afghan Cultural Information Crowdsourcing Platform

3.14 System Implementation

The backend implementation follows the modular NestJS architecture described earlier. The authentication module manages registration, login, logout, refresh sessions, password reset, email verification, and the supported Google and Facebook authentication flows. The users module manages profiles, roles, account status, and suspension. Authentication is global by default, and protected operations additionally use role, verified-email, ownership, and workflow checks where necessary. The backend issues its own access and refresh tokens regardless of whether the user logged in with a password or through a supported OAuth provider.

The entries module manages Cultural Entry drafts, editing, submission, stable keys, Persian public slugs, sources, tags, internal links, content versions, and public retrieval. The taxonomy module manages provinces, districts, categories, content types, and tags. The media module handles image validation, Cloudinary uploads, thumbnails, and YouTube link parsing. The moderation module manages the pending-submission queue and the approve, request-changes, reject, hide, restore, and archive actions. Separate modules handle bookmarks, ratings, public reviews, correction suggestions, reports, audit records, and administrative functions.

The Cultural Entry workflow is controlled by status rules rather than unrestricted updates. A new entry begins as DRAFT and can be submitted to PENDING_REVIEW. A moderator may publish it through approval, return it as CHANGES_REQUESTED, or mark it REJECTED. An entry returned for changes can be edited and resubmitted. Published content may later be hidden, restored, or archived. Important submissions and published modifications create permanent ContentVersion records. Accepted corrections and administrator updates therefore preserve the previous content state instead of replacing it without history.

The frontend implementation follows a feature-based Next.js structure. Public pages focus on discovery and reading, while authenticated pages provide contribution, profile, bookmark, rating, review, correction, and report functions. Moderator pages expose review queues and moderation tasks, and administrator pages expose user and taxonomy management together with platform oversight. The complete interface is Persian and right-to-left. Long cultural articles use a restrained reading width and clear visual hierarchy, while forms use consistent labels, validation messages, and responsive layouts. Public search uses normalized Persian text so that common Arabic and Persian character variations, spaces, half-spaces, and selected diacritical differences do not unnecessarily prevent relevant results from being found.

3.15 Security Implementation

Security is a central part of the platform because the application stores user accounts, accepts public contributions and uploaded images, and provides moderation and administrative functions. Local passwords are hashed with Argon2id and never stored as plain text. OAuth-only accounts may exist without a local password. JWT access tokens are short-lived, while refresh tokens are kept in secure HTTP-only cookies, stored as hashes in the database, rotated during refresh, and revoked after logout or account suspension. Email-verification and password-reset tokens are also stored in hashed form, are time-limited, and are designed for single use.

Authorization is enforced by the NestJS backend rather than relying on the frontend. Protected routes require authentication unless they are explicitly marked public. Role guards restrict moderator and administrator operations, verified-email checks protect contribution-related actions, ownership checks protect user drafts and personal records, and workflow checks prevent invalid state changes. A moderator cannot approve their own Cultural Entry, suspended users are rejected centrally, and only authorized roles can resolve reports, accept corrections, manage users, or perform administrative actions.

Validation is applied at multiple levels. Zod provides immediate frontend validation, NestJS uses DTO validation through class-validator and class-transformer, and Prisma together with PostgreSQL enforces database constraints. The backend rejects unsupported request fields, controls production error messages, and avoids exposing stack traces or sensitive values. Image uploads are restricted by file type, size, and count before Cloudinary storage, and arbitrary iframe code or unrestricted executable HTML is not accepted for YouTube or Tiptap content.

Additional protection is provided through Helmet security headers, controlled CORS configuration, NestJS rate limiting, environment-based secret management, application logging, and business audit records. Stronger rate limits are applied to operations that are more vulnerable to repeated abuse, such as login, registration, password reset, token refresh, reports, public reviews, correction suggestions, and image uploads. Sensitive values including database credentials, JWT secrets, OAuth credentials, Cloudinary secrets, and SMTP passwords are kept outside the source code. Important moderation and administrative actions are recorded so that changes to published cultural information and account permissions remain traceable.

This chapter presented the research methods, system-development process, requirements, architecture, technologies, database design, implementation, and security of the proposed platform, showing how the research findings were translated into the final software artifact.

Chapter 4:

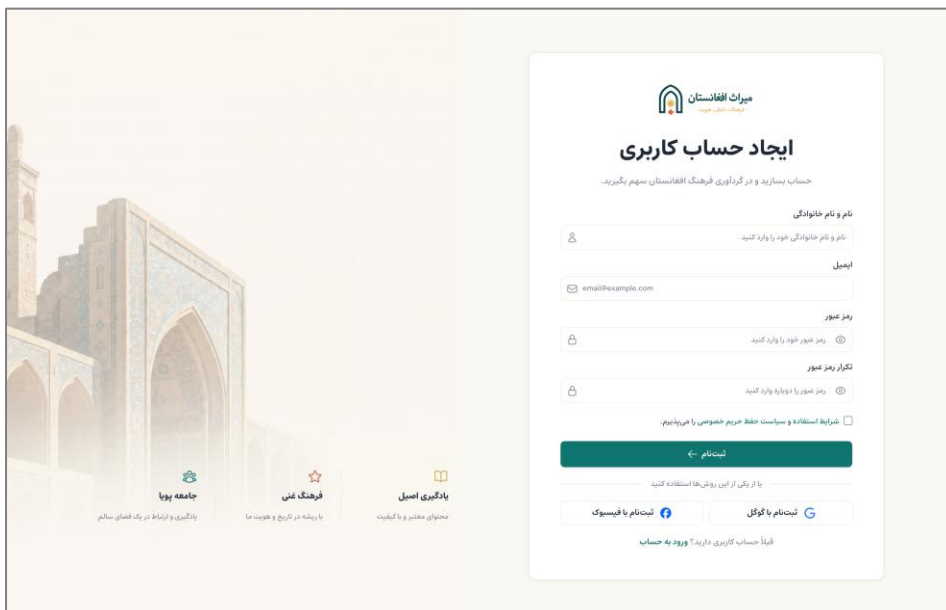
Results

4.1 Introduction

This chapter presents the completed Afghan Cultural Knowledge Platform and demonstrates the main results of the design and implementation process described in Chapter 3. The purpose of the chapter is to show how the requirements derived from the literature and system design were translated into a working web application. The chapter therefore focuses on the final system structure, the principal public and authenticated workflows, and representative interfaces rather than on source code or low-level implementation details.

The platform contains many screens and states, and several operations are performed through multi-step interfaces. Printing every page would add repetition without improving the demonstration of the system. The figures in this chapter are therefore selected to represent the major functional areas: access and roles, cultural-information discovery, presentation of cultural entries, contribution, moderation, community participation, user contribution management, moderator work, and administration. Where a workflow contains several important stages, the relevant stages are shown separately so that the sequence can be followed clearly.

4.2 Access and User Roles



The image shows a user registration interface for the 'Miran Afghanistan' website. The background features a large, ornate illustration of a traditional Afghan building with a minaret. The interface is divided into two main sections: a left sidebar with navigation links and a right main content area.

Left Sidebar Navigation:

- جامعه پویا (Dynamic Community) - با راهکارهای در یک فضای سالم (With healthy solutions in a healthy space)
- فرهنگ نئی (New Culture) - با پیشه در تاریخ و هویت ما (With tradition in history and our identity)
- پادگویی اصیل (Authentic Storytelling) - محتوای معاصر و با کیفیت (Contemporary and high-quality content)

Main Content Area:

ایجاد حساب کاربری (Create User Account)

حساب بسازید و در گردآوری فرهنگ افغانستان سهم بگیرید. (Create an account and contribute to the collection of Afghan culture.)

نام و نام خانوادگی (Name and Surname):

ایمیل (Email):

رمز عبور (Password):

تکرار رمز عبور (Repeat Password):

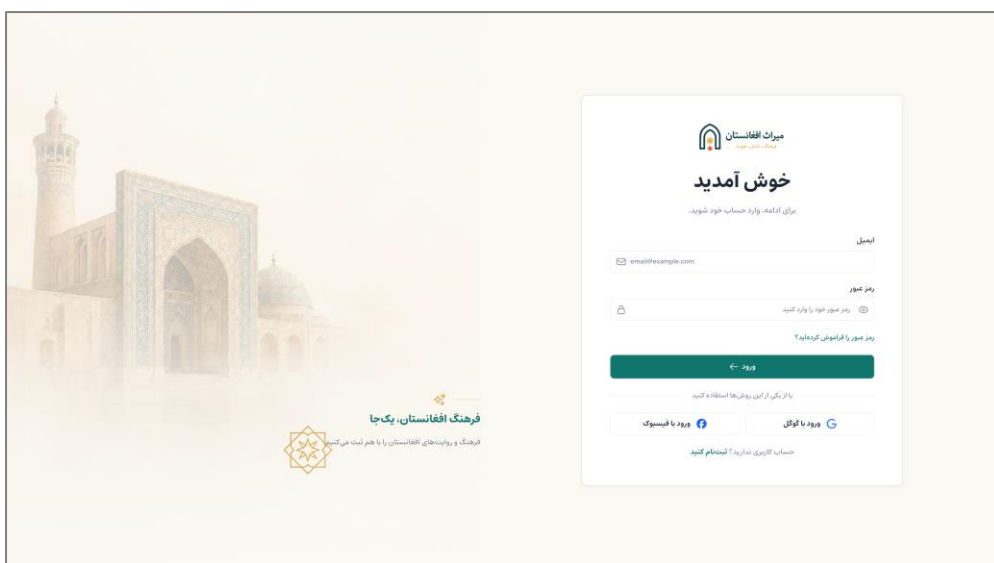
☐ شرایط استفاده و سیاست حفظ حریم خصوصی را می‌پذیریم (I agree to the Terms of Use and Privacy Policy)

ثبت نام (Register) [Green Button]

یا از یکی از این روش‌ها استفاده کنید (Or use one of these methods)

قبلاً حساب کاربری دارید؟ [ورود به حساب](#) (Already have an account? [Log in](#))

Figure 4.2.1 User Registration Interface.



The image shows a user login interface for the 'Miran Afghanistan' website. The background features a large, ornate illustration of a traditional Afghan building with a minaret. The interface is divided into two main sections: a left sidebar with navigation links and a right main content area.

Left Sidebar Navigation:

- فرهنگ افغانستان، یک جا (Afghan Culture, One Place) - فرهنگ و روایت‌های افغانستان را با هم ثبت می‌کنیم (We record Afghan culture and stories together)

Main Content Area:

خوش آمدید (Welcome)

برای ادامه، وارد حساب خود شوید. (To continue, log in to your account.)

ایمیل (Email):

رمز عبور (Password):

☐ رمز عبور را فراموشی کرده‌اید؟ (Forgot your password?)

ورود (Log In) [Green Button]

یا از یکی از این روش‌ها استفاده کنید (Or use one of these methods)

حساب کاربری ندارید؟ [ثبت نام کنید](#) (Don't have an account? [Sign up](#))

Figure 4.2.2 User Login Interface

4.3 Home Page

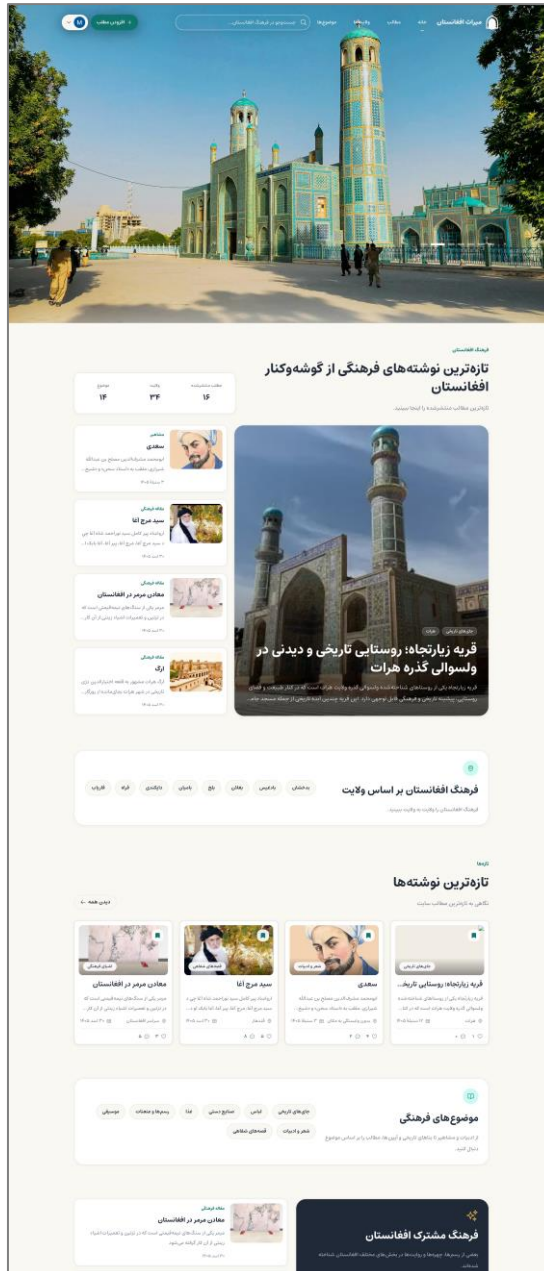


Figure 4.3.1 Home Page of the Afghan Cultural Knowledge Platform

4.4 Exploring Cultural Entries

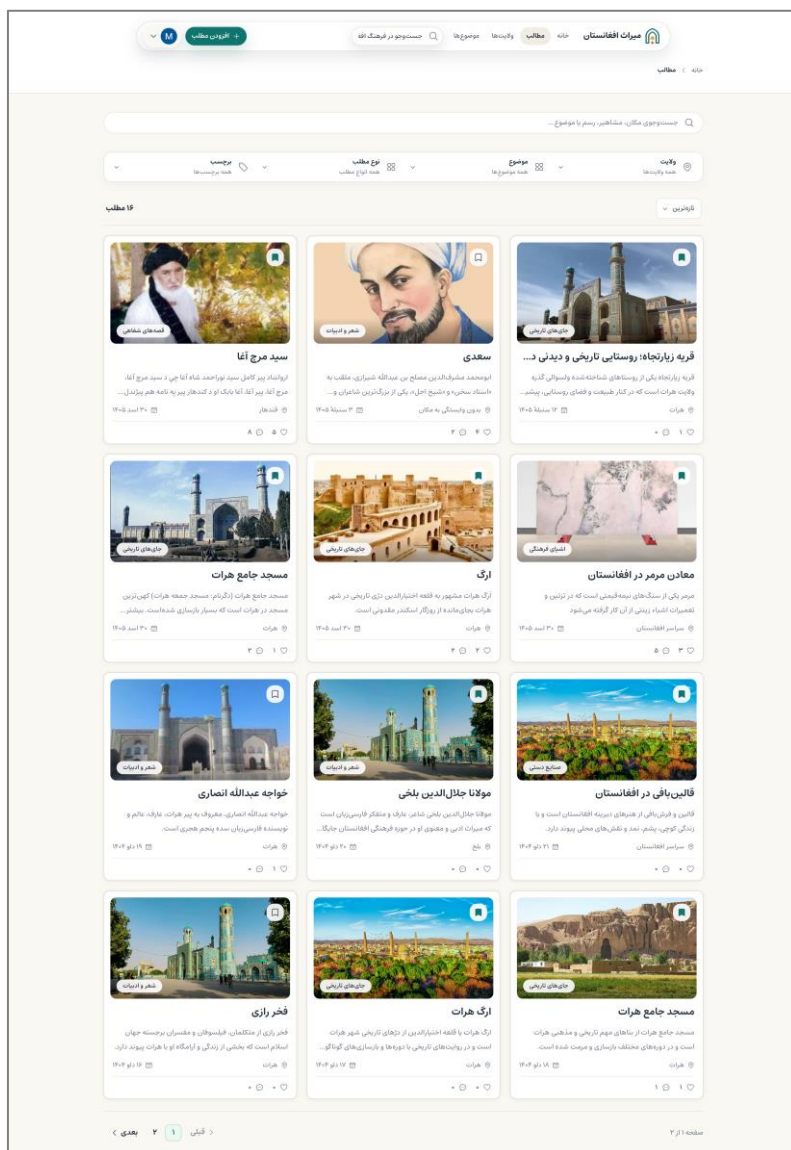


Figure 4.4.1 Cultural Content Exploration Interface

4.5 Search and Filtering

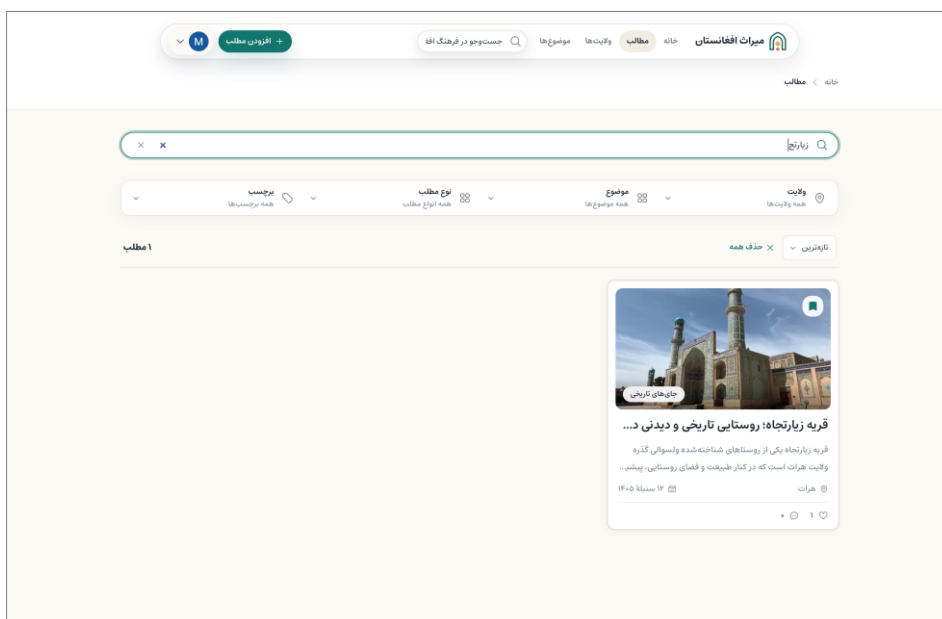


Figure 4.5.1 : Searching Cultural Entries

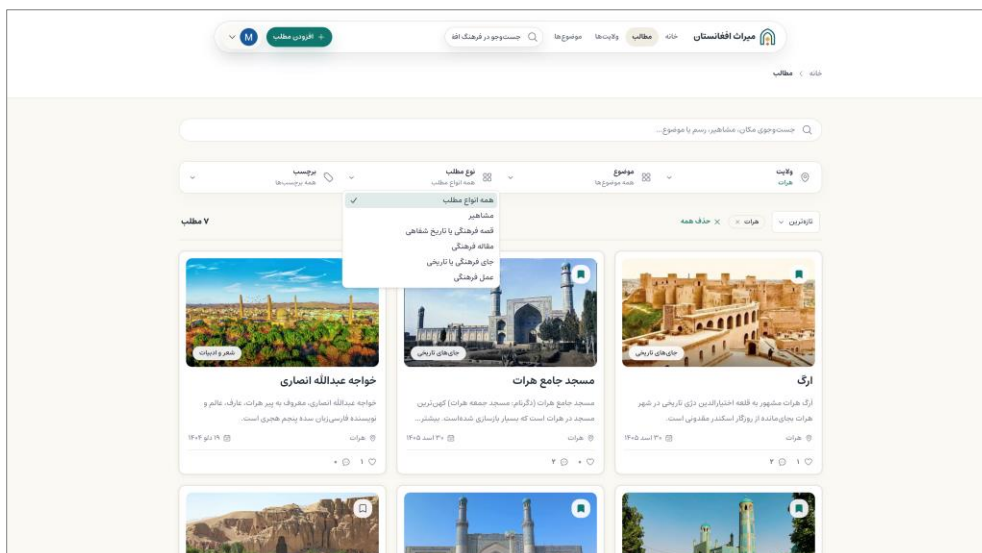


Figure 4.5.2 : Filtering Cultural Entries by Classification and Geography.

4.6 Province and Topic-Based Discovery

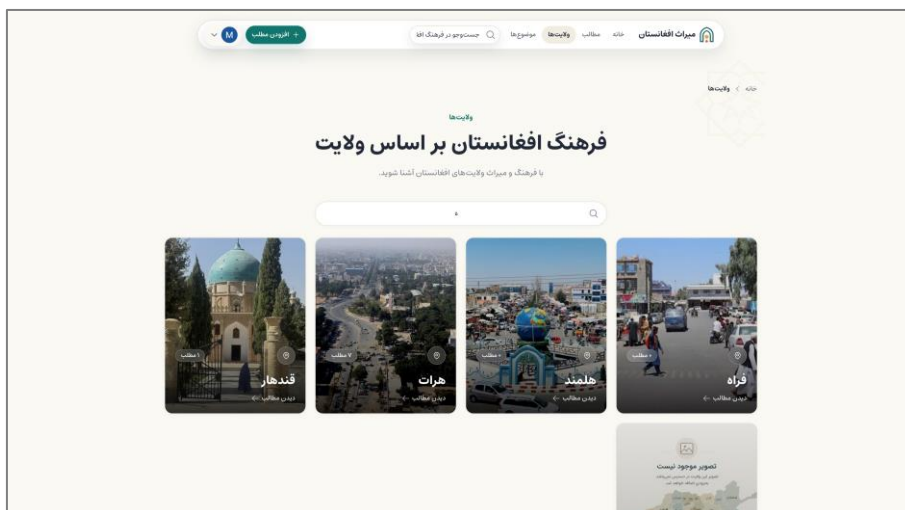


Figure 4.6.1 Provinces explore page

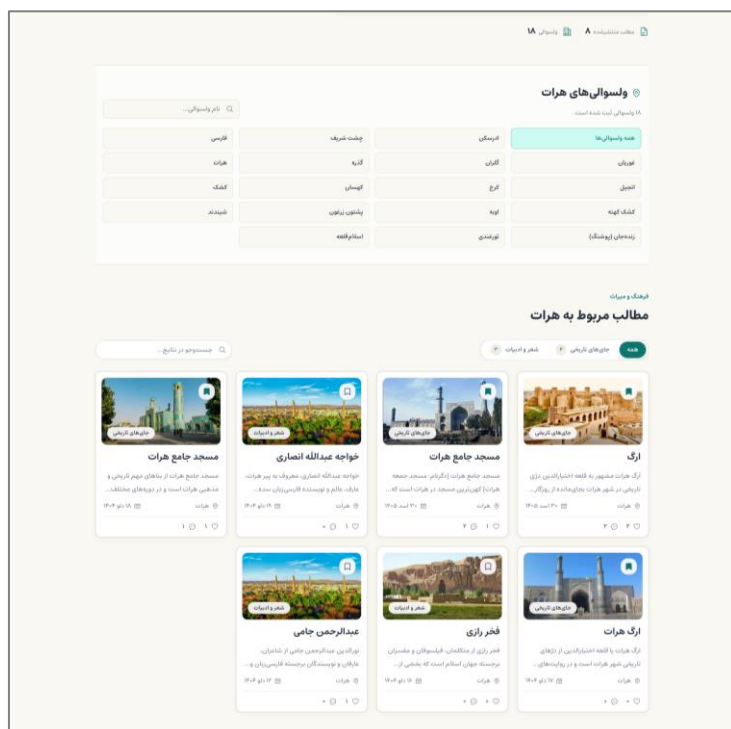


Figure 4.6.2 Province detail page with Cultural Entries Associated with the Province

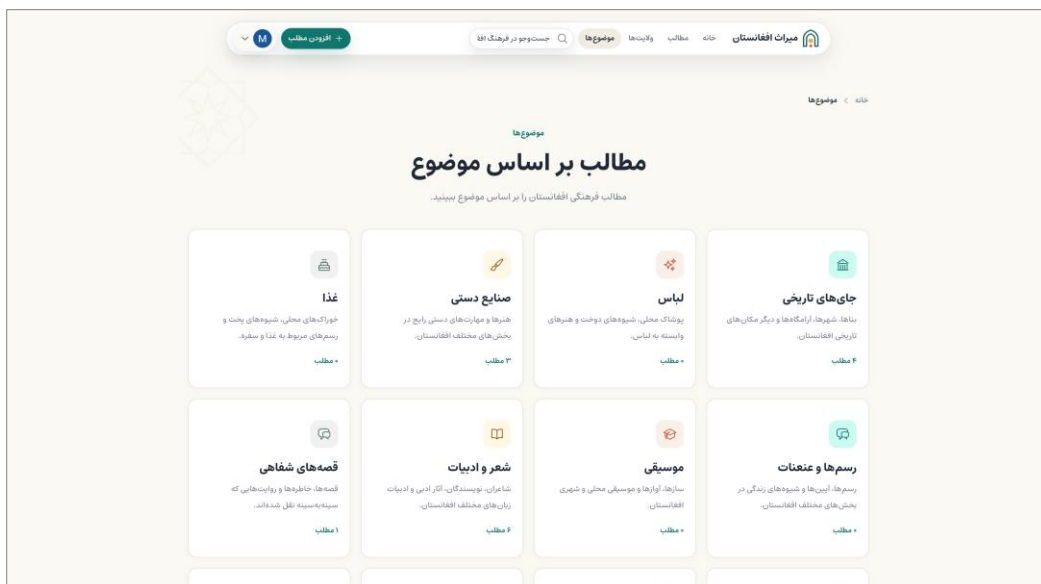


Figure 4.6.3 Topics explore page

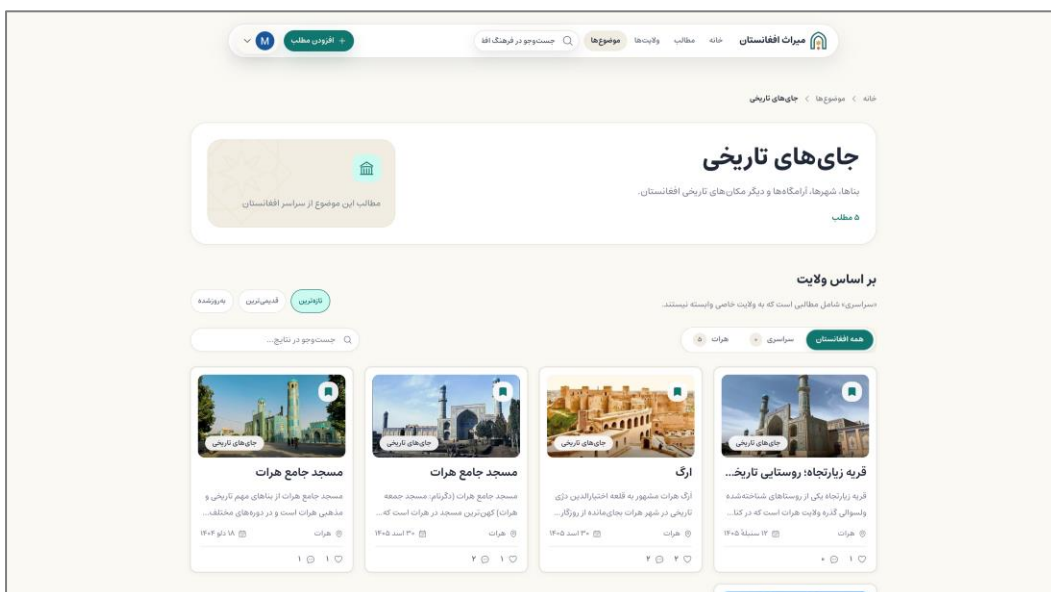


Figure 4.6.4 Topic detail page with Cultural Entries Associated with that topic.

4.7 Entry Detail Page

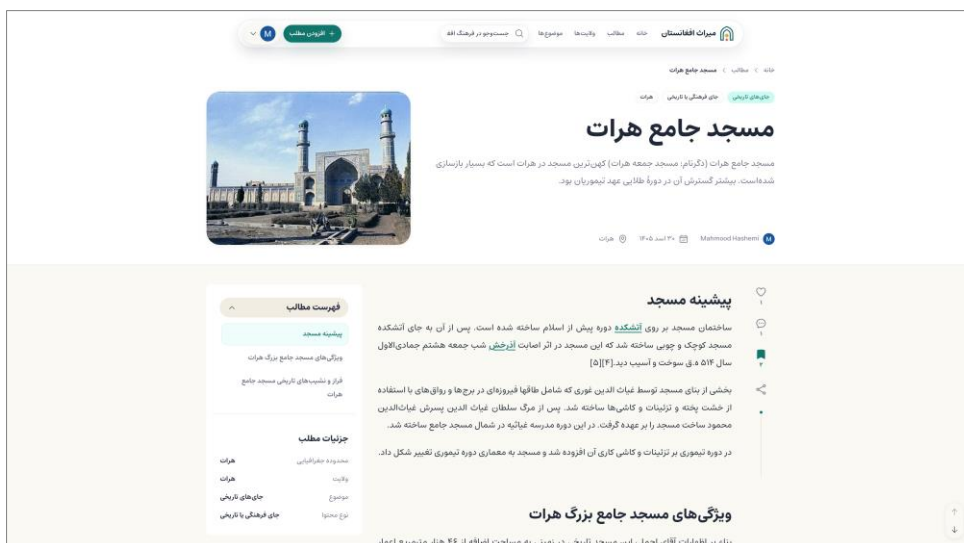


Figure 4.7.1 Published Cultural Entry Detail Page

4.8 Sources and References



Figure 4.8.1 Sources and References of a Cultural Entry

4.9 Creating a Cultural Entry

ارسال برای بررسی

ذخیره پیش نویس

پیش نمایش

مطلب جدید →

عنوان مطلب

عنوان خود را بنویسید...

از ۶۰ حرف

خلاصه

در چند جمله بگویید این مطلب درباره چیست...

از ۱۶۰ حرف

متن مطلب را بنویسید...

جزئیات مطلب

هنوز کامل نشده

▼

Figure 4.9.1 Cultural Entry Creation Interface.

[illegible][illegible]

4.10 Moderation

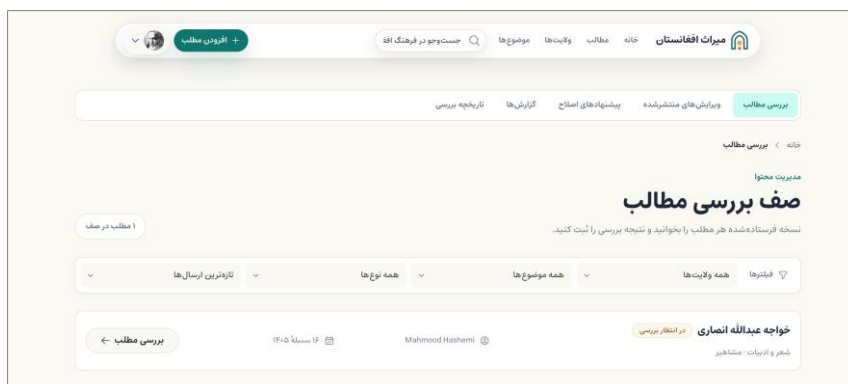


Figure 4.10.1 Moderator Review Queue



Figure 4.10.2 : Cultural Entry Review Interface



Figure 4.10.3 Moderator Decision on a Submitted Entry

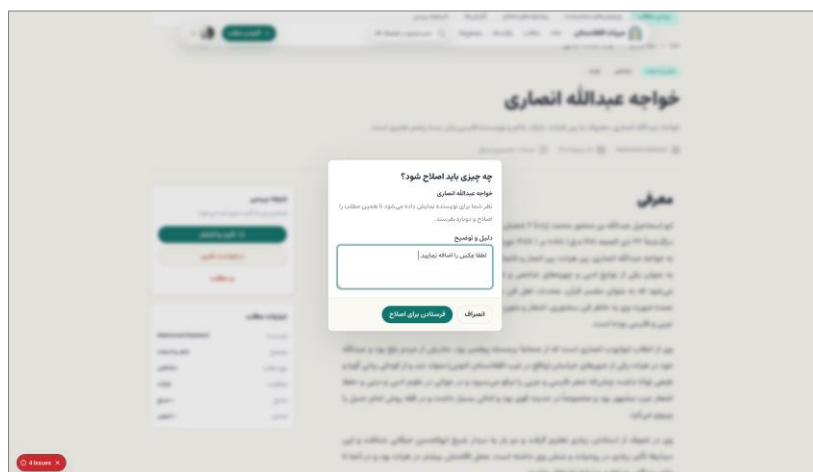


Figure 4.10.4 Entry Returned for Changes

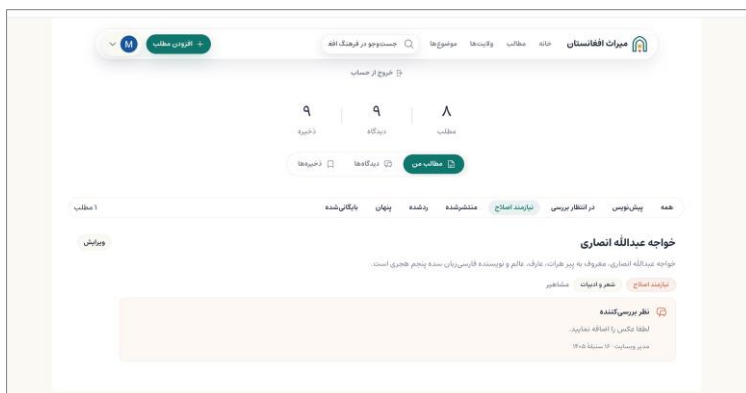


Figure 4.10.5 : Entry Prepared for Resubmission

4.11 Community Participation and Content Improvement

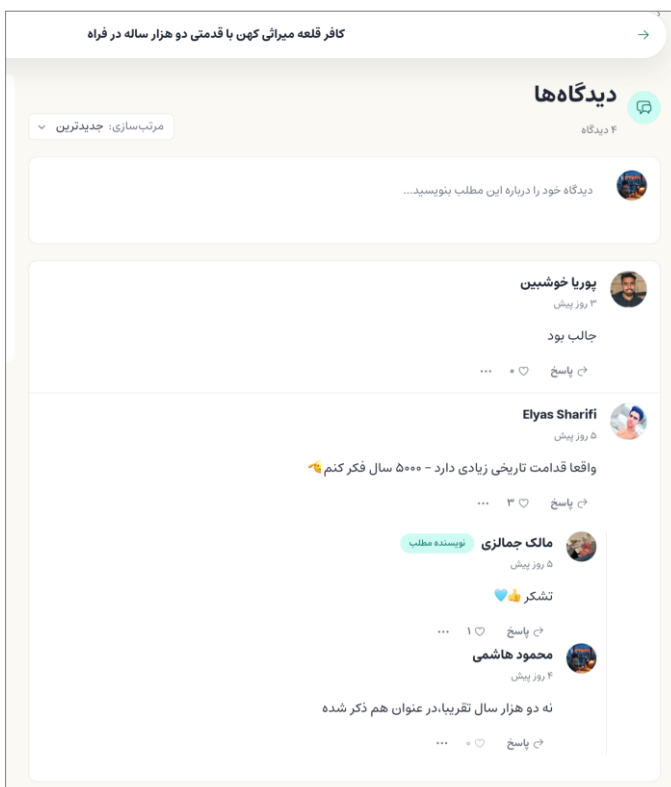


Figure 4.11.1 Comments and Replies on a Cultural Entry

براساس داستان های حم
بناهای تاریخی سیستان ب
پهلوان نامور فراه زمین یع
بر اساس افسانه های مرد
این اصل این قلعه را «کافر



چاه یا زندان کاف

ساختار «کافر قلعه» نشان

در قسمت شمالی آن چاه
استفاده می شد.

به روایت تاریخ، این یکی از
آمده است که در آن زندانیا
تاریخ بوده است.

دیواره های این چاه با خش

Figure 4.11.2 Likes and Bookmarks on Published Cultural Content

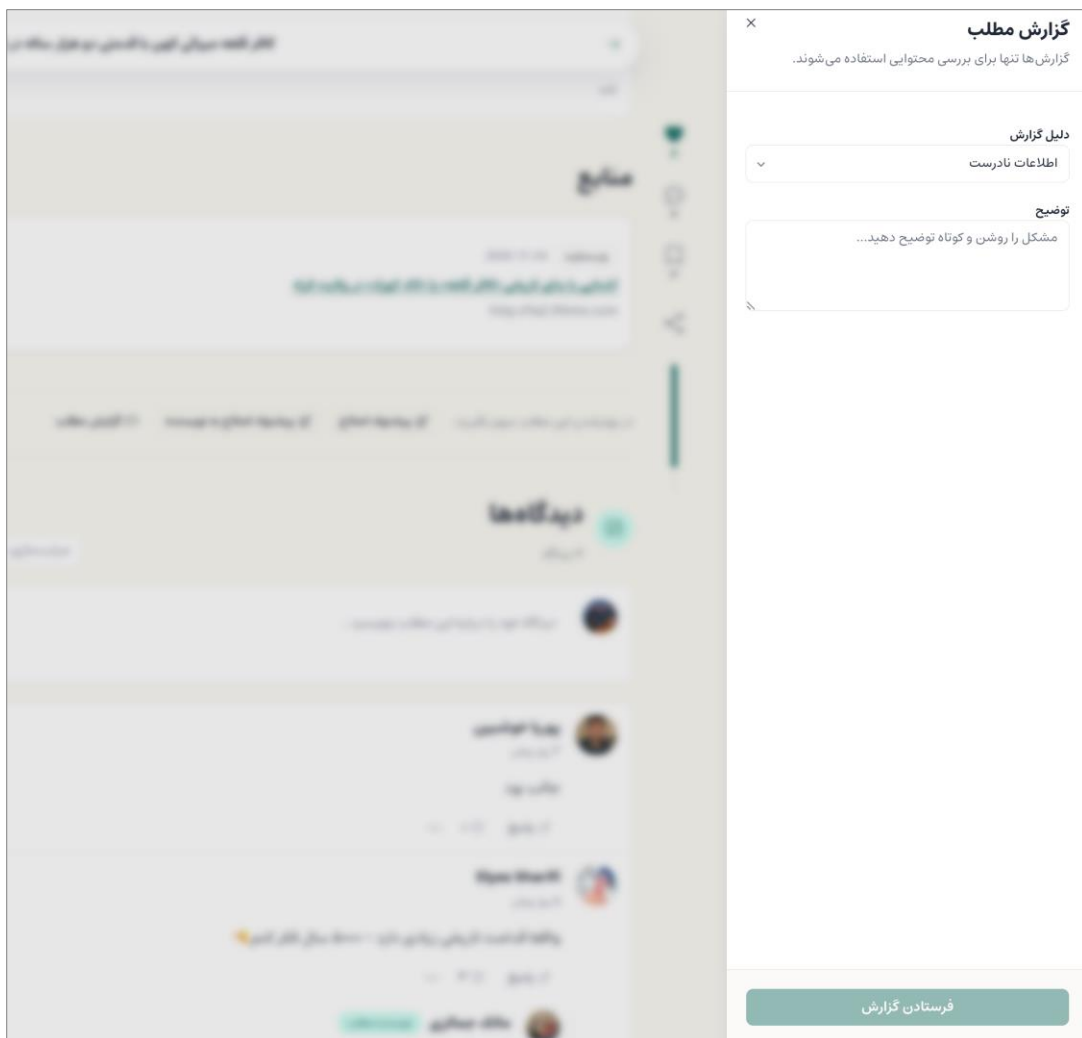


Figure 4.11.4 Reporting Problematic Cultural Content.

4.12 User Profile

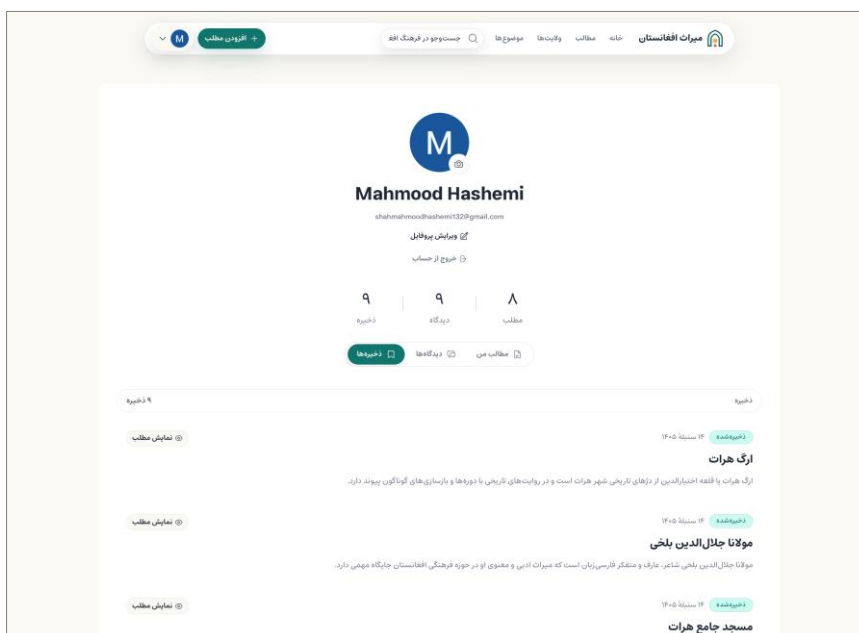


Figure 4.12.1 User Profile

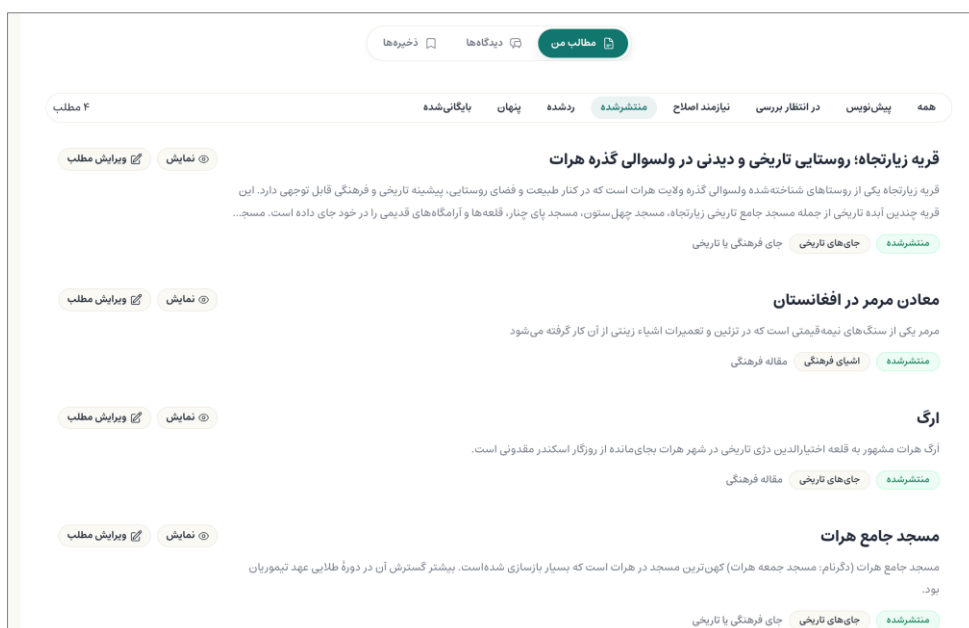


Figure 4.12.2 Status of User Cultural Contributions

4.13 Moderator Workspace

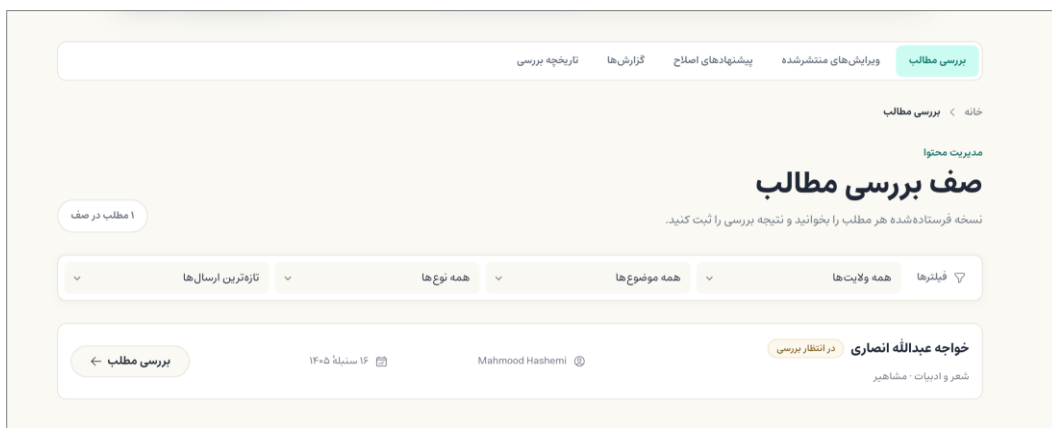


Figure 4.13.1 Moderator Workspace

4.14 Administrator Dashboard

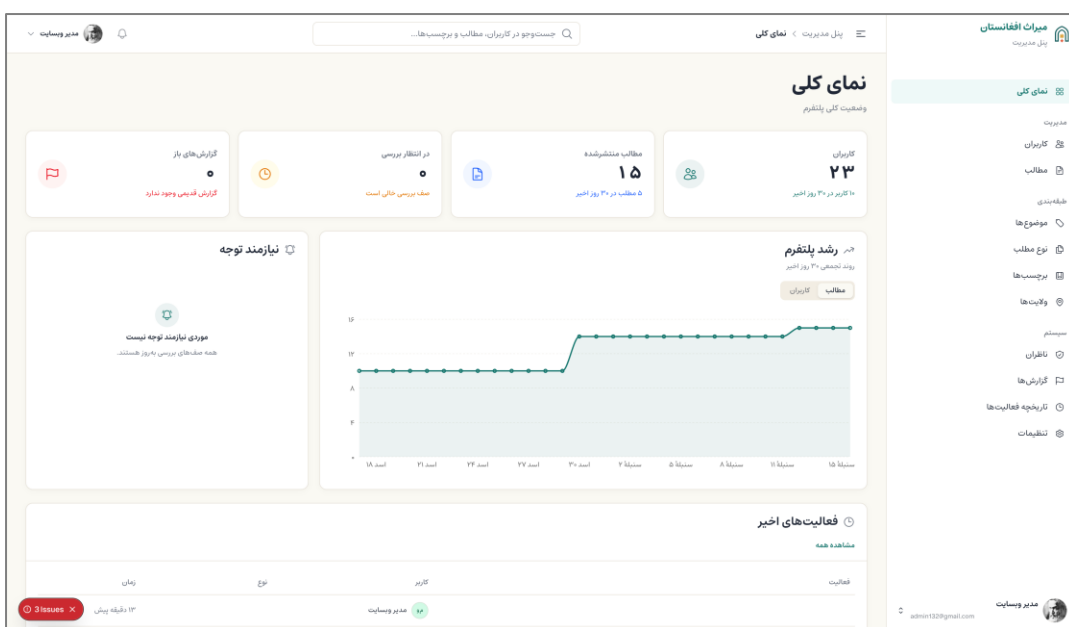


Figure 4.14.1 Administrator Dashboard.

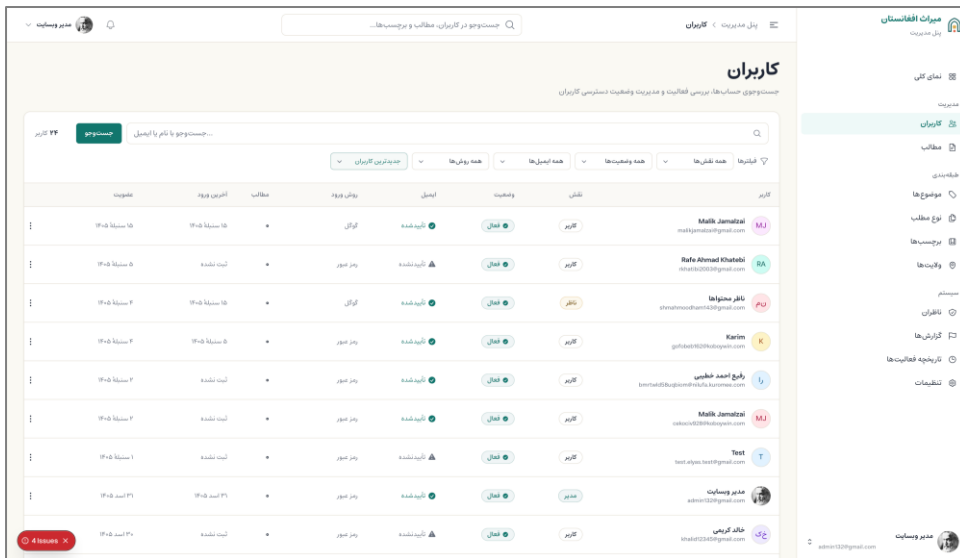


Figure 4.14.2 User Role Management

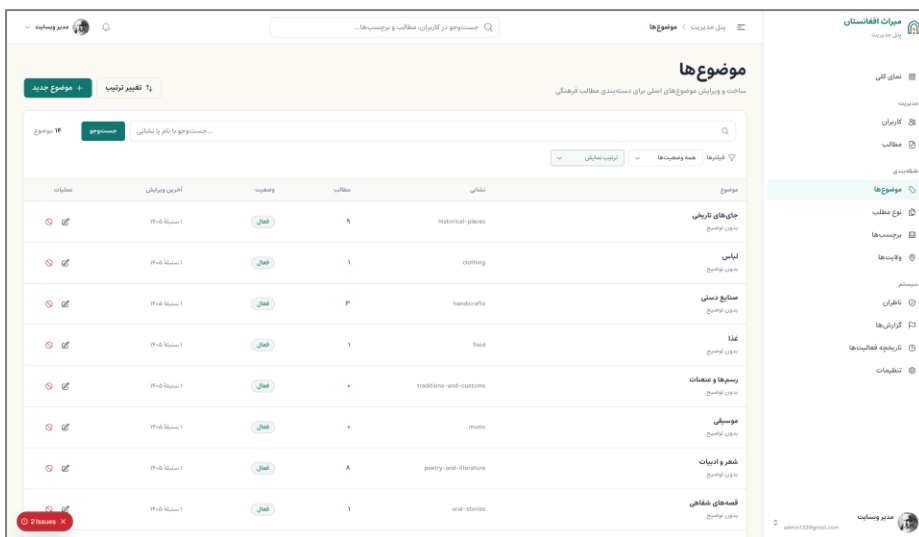


Figure 4.14.3 Topic Management

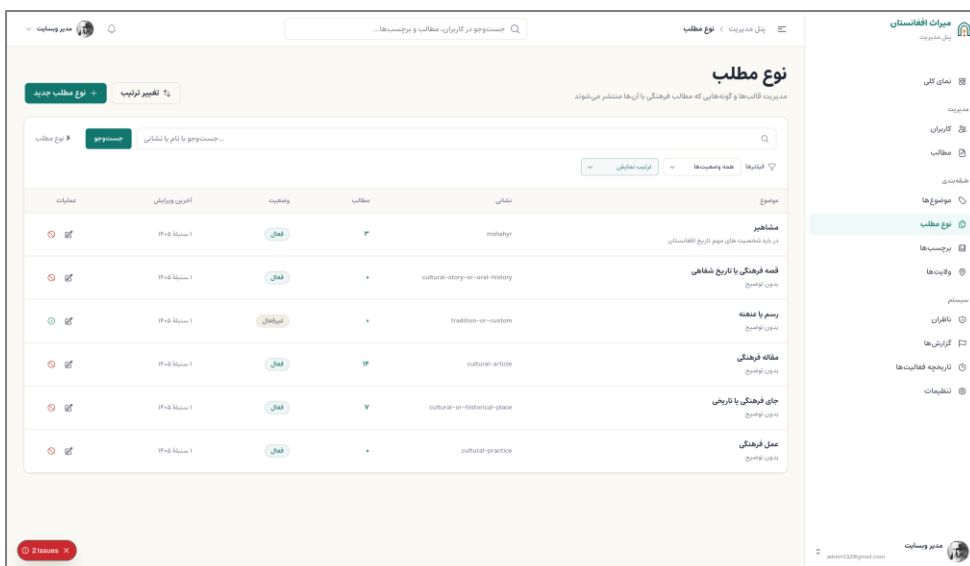


Figure 4.14.4 Content type Management

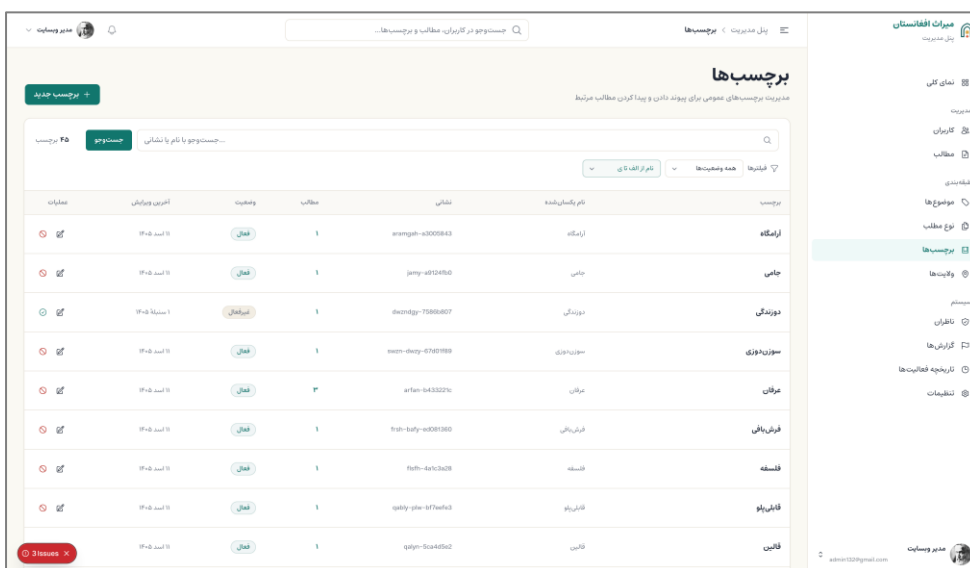


Figure 4.14.5 Tags Management

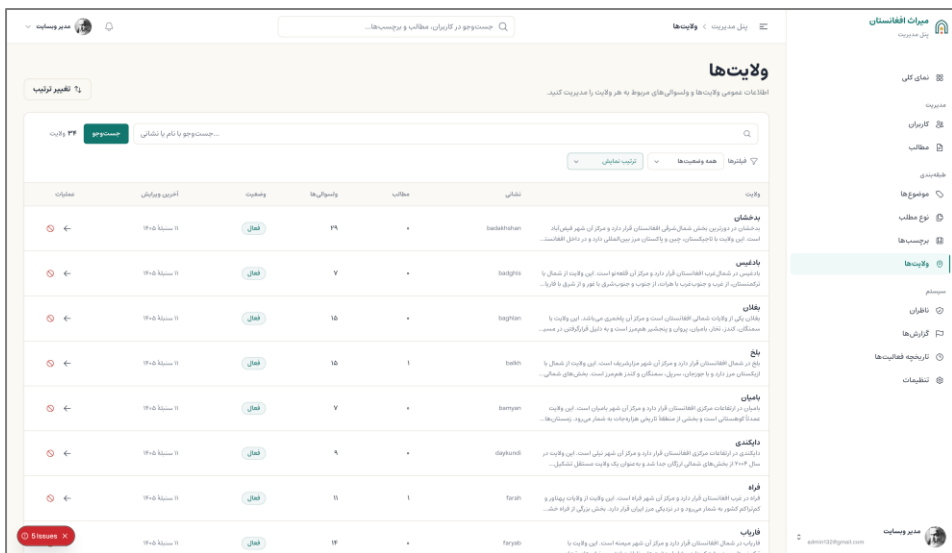


Figure 4.14.6 Provinces Management

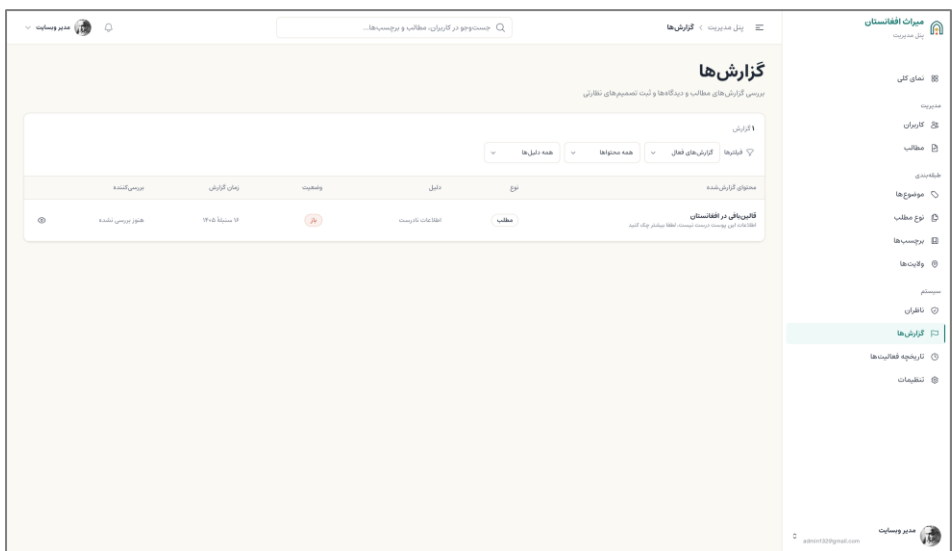


Figure 4.14.7 Reports Management

Chapter 5:

Discussion

5.1 Introduction

This chapter discusses the results presented in Chapter 4 in relation to the research problem, research objectives, and the requirements identified during the study. Chapter 4 demonstrated the completed Afghan Cultural Knowledge Platform through representative interfaces covering access and roles, cultural-information discovery, presentation of cultural entries, contribution, moderation, community participation, user contributions, moderator work, and administration. The purpose of this chapter is therefore not to repeat the screenshots, but to explain what the implemented features mean in relation to the challenges of accessing, organizing, contributing to, and maintaining Afghan cultural information. The discussion follows the same order as Chapter 4 so that each major result can be considered in its functional and research context.

5.2 Access and User Roles

Figure 4.2.1 and *Figure 4.2.2* demonstrate the registration and login interfaces of the platform. These interfaces provide the basic authentication mechanism through which users access the system and become identified participants. User accounts allow contributions to be associated with individual users, supporting the management of submitted content, communication with contributors during the review process, and the maintenance of user contribution records. The authentication mechanism therefore establishes the basic access structure required for participation in the platform.

The platform organizes participation through three main roles: **registered users, moderators, and administrators**. Registered users can discover cultural information and contribute new cultural content to the platform. Moderators are responsible for reviewing submitted contributions and handling the review process, including approving, rejecting, or requesting changes to submitted content. Administrators are responsible for broader platform management, including user and role management, taxonomy management, and monitoring reports and platform activity. These roles define the responsibilities and permissions associated with different types of participants.

Role separation also provides a basis for managing the different functions of the platform. Registered users participate in the creation and improvement of cultural information, while moderation functions are assigned to moderators and broader management functions are assigned to administrators. Associating these functions with specific roles separates content contribution, content review, and platform administration. The resulting access structure supports the operation of the platform as a managed cultural knowledge environment

in which users can participate while review and administrative activities remain assigned to their respective roles.

5.3 Cultural Information Discovery

Figure 4.3.1 demonstrates the home page of the Afghan Cultural Knowledge Platform. The home page provides the main entry point to the platform and presents users with access to the available cultural information and exploration features. It provides an overview of the platform and guides users toward discovering cultural entries through the available navigation and content sections. This interface establishes the initial structure through which cultural information is presented and accessed within the platform.

Figure 4.4.1 demonstrates the exploration of cultural entries. The interface presents cultural records as an organized collection that users can browse to discover available information. Each entry is presented as an individual cultural record within the platform, allowing users to navigate through the available content. This functionality supports the organization and accessibility of cultural information by providing a consistent method for browsing the collected entries.

Figure 4.5.1 and *Figure 4.5.2* demonstrate the search and filtering functionality of the platform. The search function allows users to locate cultural entries using relevant search terms, while the filtering functions allow the available entries to be narrowed according to classification and geographical information. These functions provide users with multiple ways to locate cultural information based on their specific interests or requirements. The combination of search and filtering therefore supports the discovery and navigation of cultural entries within the platform.

Figure 4.6.1 and *Figure 4.6.2* demonstrate province-based discovery, while *Figure 4.6.3* and *Figure 4.6.4* demonstrate topic-based discovery. Province-based discovery organizes cultural entries according to their geographical context, allowing users to explore information associated with a particular province. Topic-based discovery organizes entries according to shared cultural subjects or classifications, allowing users to explore related information through a specific topic. These features provide additional navigation paths for accessing the cultural information available on the platform and demonstrate the organization of cultural entries according to relevant geographical and thematic relationships.

5.4 Cultural Entry Presentation and Sources

Figure 4.7.1 demonstrates the published cultural entry detail page. The page presents the information associated with a cultural entry in a structured format.

It includes the cultural title, descriptive content, classification, geographical context, and supporting information associated with the entry. These elements provide the main details required to describe and contextualize a cultural record within the platform. The entry detail page therefore provides a consistent structure for presenting individual cultural records.

Figure 4.8.1 demonstrates the sources and references associated with a cultural entry. The interface provides a dedicated area for recording the sources used to support the submitted cultural information. These references are associated with the corresponding entry and are available as part of the entry's information. Including sources within the cultural entry also supports the review process by providing moderators with additional information to consider when evaluating submitted content.

The combination of structured entry information and associated sources provides a consistent format for storing and presenting cultural records. Cultural entries can therefore contain not only descriptive information, but also their classification, geographical context, and supporting references. This structure establishes the information presentation model used by the platform for organizing and displaying cultural knowledge.

5.5 Cultural Content Contribution Workflow

Figure 4.9.1 demonstrates the interface for creating a cultural entry. The interface provides registered users with a mechanism for submitting new cultural information to the platform. Through this functionality, users can initiate a cultural entry and provide information that can subsequently proceed through the platform's contribution and review workflow. This establishes the user contribution component of the crowdsourcing model and allows content creation to be carried out by registered participants.

Figure 4.9.2 demonstrates the addition of cultural content and classification information. At this stage, the contributor provides the descriptive information for the cultural entry together with its relevant classification and metadata. The classification information associates the entry with the platform's organizational structure, while the descriptive content provides the main information to be presented as part of the cultural record. Including these elements during the contribution process establishes a consistent structure for submitted cultural entries.

Figure 4.9.3 demonstrates the addition of sources and supporting media. Contributors can provide references associated with the cultural information and add supporting media to the entry. These elements provide additional context for the submitted content and form part of the information available

for review. Sources can also assist moderators in examining the basis of a contribution before making a publication decision.

The three stages shown in Chapter 4 represent the contribution workflow implemented by the platform. The process begins with the creation of a cultural entry, continues with the addition of content and classification information, and includes supporting sources and media before the contribution proceeds to review. This workflow provides a structured mechanism for community members to contribute cultural information while keeping submitted content within the platform's review process before publication.

5.6 Moderation and Publication Workflow

The moderation results shown in *Figure 4.10.1* to *Figure 4.10.2* demonstrate the review stage of the crowdsourcing process. *Figure 4.10.1* demonstrates the moderator review queue, where submitted cultural entries are presented for review. *Figure 4.10.2* demonstrates the interface used to inspect an individual submitted entry and its associated information. Together, these interfaces establish a separate review stage between the submission of cultural content and its publication on the platform.

Figure 4.10.3 demonstrates the moderation decision stage. After inspecting a submitted entry, the moderator can make a decision regarding its status within the review process. This establishes a defined transition between submitted and published cultural information. The moderation decision therefore forms an essential part of the content workflow by determining the outcome of a submitted contribution before it becomes part of the published cultural collection.

Figure 4.10.4 and *Figure 4.10.5* demonstrate the requested-changes and resubmission workflow. The requested-changes state indicates that additional modification is required before the entry can proceed, while the resubmission interface allows the contributor to revise the entry and submit it again for review. This creates an iterative workflow between contributors and moderators, allowing submitted content to move through review, revision, and subsequent resubmission when necessary.

Overall, the figures in this section demonstrate the moderation workflow implemented by the platform: **submitted content** → **moderator review** → **moderation decision** → **requested changes and resubmission when required** → **subsequent review**. This workflow separates contribution from publication and provides the platform with a structured process for reviewing and managing submitted cultural information.

5.7 Community Participation and Content Improvement

Figure 4.11.1 demonstrates comments and replies on a cultural entry. This feature allows users to participate in discussions related to published cultural information. Users can add comments to an entry and respond to existing comments, providing a space for discussion around the information presented. This extends community participation beyond the submission of new cultural entries and allows interaction to take place around published content.

Figure 4.11.2 demonstrates likes and bookmarks. The like function allows users to interact with published cultural entries, while the bookmark function allows users to save entries for later access. These functions provide additional forms of user interaction with cultural content beyond viewing the information. They therefore form part of the platform's community participation features.

Figure 4.11.3 demonstrates suggested corrections to published cultural information. This feature allows users to identify information that they believe requires modification and submit a correction suggestion for the corresponding entry. Suggested corrections provide a mechanism for community members to contribute to the improvement of existing cultural records rather than limiting contributions to the creation of new entries. The suggested information can then be handled through the platform's content-management and review processes.

Figure 4.11.4 demonstrates the reporting functionality for problematic cultural content. Through this feature, users can report published content that they consider problematic, allowing the issue to be brought to the attention of the platform's responsible users. Reporting therefore provides a mechanism for identifying issues in published content after the initial publication process.

Together, the features demonstrated in this section provide several forms of community participation, including **discussion, interaction, bookmarking, correction suggestions, and reporting**. These functions extend participation beyond the initial contribution and moderation stages and provide mechanisms through which users can interact with, provide feedback on, and raise concerns about published cultural information.

5.8 User Profile and Personal Contributions

Figure 4.12.1 demonstrates the user profile, while *Figure 4.12.2* demonstrates the status of the user's cultural contributions. The user profile provides a dedicated view of the user's information and participation within the platform. The contribution-status view presents the cultural entries associated with the user together with their current states in the contribution workflow. This

allows contributors to view the information they have submitted and follow its progress through the platform's review process.

The contribution states demonstrated in Chapter 4 include **Draft, Pending Review, Published, Changes Requested, and Rejected**. These states represent the different stages and outcomes that a cultural contribution can have within the platform. A draft represents content that has not yet been submitted for review, while a pending-review contribution has been submitted and is awaiting moderation. Published, changes-requested, and rejected states indicate the subsequent outcomes of the moderation process. The status model therefore provides contributors with information about the current state of their submissions and establishes a clear connection between user contributions and the platform's moderation workflow.

Overall, the user profile and contribution-status features provide users with a dedicated view of their participation and submitted cultural content. Together with the contribution and moderation workflows discussed in the previous sections, these features complete the user-side contribution cycle from **creating and submitting content to reviewing its current status and outcome**.

5.9 Moderator Workspace

Figure 4.13.1 demonstrates the moderator workspace. The workspace provides a dedicated area for managing submitted cultural contributions and carrying out moderation activities. It brings together the functions required for reviewing submitted entries, allowing moderators to identify submissions, inspect their content, and process them according to the moderation workflow. The workspace therefore provides a centralized interface for carrying out the review activities presented in the previous section.

The moderator workspace also connects the contribution and moderation stages of the platform. After a contributor submits a cultural entry, the submission becomes available within the moderation environment for further review. Moderators can use this workspace to examine submitted information and manage its progress through the available moderation actions. This establishes a defined operational area for content review within the platform and separates moderation activities from the user-facing cultural information interfaces.

Overall, the moderator workspace provides the operational interface through which moderators manage submitted cultural information. Together with the review queue, entry inspection, moderation decisions, and requested-changes workflow discussed earlier, it forms the moderator-side component of the platform's contribution and publication process.

5.10 Administration and Platform Governance

Figure 4.14.1 demonstrates the administrator dashboard. The dashboard provides a central interface for accessing and managing the administrative functions of the platform. It presents an administrative view that is separate from the interfaces used for discovering and contributing cultural information. The dashboard therefore serves as the main entry point for administrators to access the management functions of the system.

Figure 4.14.2 demonstrates user role management. This function allows administrators to manage the roles assigned to participants within the platform. The role-management interface corresponds to the role structure described earlier, in which registered users, moderators, and administrators have different responsibilities and permissions. It therefore provides an administrative mechanism for managing participant roles within the system.

Figure 4.14.3 to *Figure 4.14.6* demonstrate taxonomy management for topics, content types, tags, and provinces. These interfaces allow administrators to manage the classification elements used to organize cultural entries. The taxonomy functions are connected to the information structure of the platform because the classifications are used when organizing and discovering cultural information. Administrative control over these elements provides a mechanism for maintaining the classification structure used by cultural entries.

Figure 4.14.7 and *Figure 4.14.8* demonstrate reports management and administrative activity and oversight. These functions provide administrators with access to reported issues and information about platform activity. They extend the administrative functions beyond user and taxonomy management by providing mechanisms for reviewing reported content and monitoring activities within the platform.

Overall, the administrative features demonstrated in this section provide functions for **dashboard management, user and role management, taxonomy management, reports management, and activity oversight**. Together with the contribution, moderation, and community features discussed in the previous sections, these functions establish the administrative component of the platform and complete the main management structure of the implemented system.

Chapter 6:

Conclusion

6.1 Introduction

This chapter concludes the study by returning to the research objectives and research questions defined at the beginning of the research. Chapter 4 presented the implemented Afghan Cultural Knowledge Platform, while Chapter 5 discussed how the implemented features address the identified challenges. This chapter summarizes the achievement of the research objectives, answers the research questions, discusses the conditions under which the study was carried out, and provides concluding remarks about the developed platform.

This chapter does not introduce new evidence. Instead, it brings together the findings from previous chapters and explains what was achieved through the design, development, and demonstration of the proposed solution.

6.2 Achievement of the Research Objectives

The study was conducted based on one main objective and three sub-objectives. The achievement of each objective is discussed below based on the findings and implementation presented throughout the research.

6.2.1 To Identify the Challenges Affecting Access to Afghan Cultural Information

This objective was achieved through the literature review and analysis presented in Chapter 2. The reviewed materials identified several challenges affecting cultural information access, including fragmented information sources, limited digital documentation, lack of structured organization, and difficulties in maintaining accessible and reliable cultural knowledge.

The identified challenges provided the foundation for defining the requirements of the proposed platform and guided the design decisions throughout the development process.

6.2.2 To Analyze the Identified Challenges and Their Effects on Access to Afghan Cultural Information

This objective was achieved through examining how the identified issues influence the availability, discoverability, organization, and reliability of cultural information. The analysis showed that when cultural knowledge remains distributed across different sources without structured organization, users face difficulties in finding, understanding, and preserving such information.

The findings from this analysis directly influenced the design of the platform, particularly the use of structured cultural entries, classification, geographical organization, references, and community-based improvement mechanisms.

6.2.3 To Develop a Crowdsourced Web-Based Solution That Addresses the Identified Challenges of Accessing Afghan Cultural Information

This objective was achieved through the design, implementation, and demonstration of the Afghan Cultural Knowledge Platform. The developed system provides mechanisms for discovering cultural information, creating and submitting cultural entries, reviewing contributions, managing references, supporting community interaction, and organizing information through categories and geographical structures.

The implemented moderation workflow also addresses reliability concerns by introducing a review process before submitted content becomes publicly available.

6.3 Achievement of the Main Objective

The main objective of this study was to investigate and analyze the challenges of accessing Afghan cultural information and develop a crowdsourced web-based solution to address these challenges.

This objective was achieved through combining the findings from the literature review with the development of the proposed platform. The research identified access-related challenges and translated them into functional requirements and system features.

The developed platform provides a structured environment where cultural information can be documented, organized, discovered, reviewed, and improved through community participation. However, the study demonstrates the developed solution as a software artifact and does not claim measurable improvements in cultural preservation or user participation through long-term public deployment.

6.4 Comparison with Traditional Cultural Documentation Approaches

Traditional cultural documentation approaches often rely on books, separate digital resources, personal records, and local knowledge. Although these approaches preserve valuable information, they may provide limited opportunities for structured organization, discovery, and continuous improvement.

The developed platform introduces a centralized environment where cultural information can be connected with categories, locations, references, and community contributions. Unlike static documentation methods, the platform provides workflows that allow information to be created, reviewed, updated, and maintained.

This comparison is based on the capabilities demonstrated by the developed system and does not represent a measurement from a large-scale deployment.

6.5 Answers to the Research Questions

The main research question of this study was:

How can the challenges of accessing Afghan cultural information be addressed through a crowdsourced web-based solution?

The study answers this question by demonstrating that a crowdsourced web-based platform can address these challenges by combining structured cultural records, community contribution workflows, moderation mechanisms, references, and interaction features within a unified environment.

6.5.1 What Are the Challenges Affecting Access to Afghan Cultural Information?

The first research question was addressed through the literature review. The identified challenges include fragmented information sources, limited digital documentation, lack of structured organization, and difficulties in maintaining reliable and accessible cultural information.

6.5.2 What are the effects of the identified challenges on access to Afghan cultural information?

The second research question was addressed through the analysis of the identified challenges. The findings show that fragmented and poorly organized information makes cultural knowledge difficult to discover, compare, and preserve. The absence of structured digital systems also limits opportunities for community participation and continuous improvement.

6.5.3 How Can a Crowdsourced Web-Based Solution Be Developed to Address the Identified Challenges of Accessing Afghan Cultural Information?

The third research question was addressed through the design and implementation of the Afghan Cultural Knowledge Platform. The developed solution combines cultural information management, user contribution,

moderation, references, and community interaction to provide a structured approach for improving access to Afghan cultural information.

6.6 Conditions Under Which the Study Was Carried Out

Several conditions influenced the scope and interpretation of this study.

First, the platform was developed and evaluated as a software artifact rather than through long-term deployment with a large community of users. Therefore, the study demonstrates system functionality rather than measuring social impact.

Second, the cultural content used during development and demonstration represents examples used for evaluating system functionality. It does not represent a complete collection of Afghan cultural information.

Third, extensive large-scale testing, including comprehensive performance and security testing, was outside the scope of this study. The evaluation focused on demonstrating implemented functionality and workflows.

6.7 Concluding Remarks

Cultural information represents an important part of collective knowledge and identity. However, without appropriate digital structures, valuable cultural information may remain difficult to access, organize, and preserve.

This study investigated the challenges affecting access to Afghan cultural information and developed a crowdsourced web-based platform to address these challenges. The developed system provides structured cultural entries, discovery mechanisms, contribution workflows, moderation processes, references, and community interaction features.

The research demonstrates how modern web technologies can support cultural knowledge management by creating an environment where information can be collected, reviewed, organized, and shared. The continued growth and usefulness of such a platform depend on future adoption and ongoing participation from communities and cultural contributors.

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Terminology

Cultural Heritage: The knowledge, traditions, practices, artifacts, places, and expressions that represent the historical, social, and cultural identity of a community and are passed between generations.

Crowdsourcing: A method of collecting information, ideas, or contributions from a large group of people, usually through an online platform, to support knowledge creation and problem-solving.

Cultural Entry: A structured digital record that represents a cultural item or topic within the platform. A cultural entry may include descriptions, classifications, geographical information, media, and supporting references.

Moderator: A user role responsible for reviewing submitted cultural content, managing quality control activities, and making decisions about publication, rejection, or requested changes.

Moderation Workflow: The process through which submitted cultural content is reviewed, approved, rejected, or returned to contributors for improvement.

چکیده

معلومات فرهنگی افغانستان بخش مهمی از هویت تاریخی و دانش جمعی این کشور را تشکیل می‌دهد؛ با این حال، دسترسی به این معلومات همچنان با چالش‌هایی مواجه است، زیرا منابع فرهنگی اغلب در میان منابع مختلف پراکنده بوده، به صورت کافی مستندسازی نشده و در یک محیط دیجیتالی یکپارچه و منظم سازمان‌دهی نشده‌اند. پلتفرم‌های دیجیتالی موجود در برخی زمینه‌ها دسترسی به معلومات فرهنگی را بهبود بخشیده‌اند، اما اغلب از مشارکت جامعه در تولید محتوا، سازمان‌دهی ساختاریافته و بهبود مستمر دانش فرهنگی پشتیبانی محدودی دارند. این پژوهش با هدف بررسی و تحلیل چالش‌های مؤثر بر دسترسی به معلومات فرهنگی افغانستان و توسعه یک راه‌حل وب‌محور مبتنی بر مشارکت جمعی انجام شده است. برای بررسی مطالعات و منابع موجود، شناسایی چالش‌های مرتبط و تعیین شکاف پژوهشی از روش تحقیق کتابخانه‌ای استفاده شده و روش‌شناسی تحقیق علوم طراحی نیز چارچوب طراحی و توسعه راه‌حل پیشنهادی را فراهم کرده است. حاصل این پژوهش، پلتفرم معلومات فرهنگی افغانستان است که امکاناتی شامل ثبت ساختاریافته اطلاعات فرهنگی، سازمان‌دهی جغرافیایی و موضوعی، منابع و مراجع پشتیبان، قابلیت‌های مشارکت جامعه، مدیریت مشارکت‌های کاربران و گردش کارهای نظارت و بررسی محتوا را فراهم می‌کند. این پلتفرم به کاربران ثبت‌شده اجازه می‌دهد تا معلومات فرهنگی را از طریق یک فرایند کنترل‌شده ارسال کنند؛ به گونه‌ای که مشارکت‌های ثبت‌شده پیش از انتشار می‌توانند مورد بررسی، اصلاح و مدیریت قرار گیرند. ناظران می‌توانند محتوای ارسال‌شده را بررسی کرده و تصمیم مناسب در مورد آن اتخاذ کنند، در حالی که مدیران مسئول مدیریت کاربران، نقش‌ها، طبقه‌بندی‌ها، گزارش‌ها و سایر امور مدیریتی پلتفرم هستند. سیستم توسعه‌یافته رویکردی ساختاریافته برای جمع‌آوری، سازمان‌دهی، بررسی و ارائه معلومات فرهنگی افغانستان از طریق مشارکت جامعه ارائه می‌کند. از آنجا که این پلتفرم با یک جامعه بزرگ از کاربران مستقر و ارزیابی نشده است، ارزیابی پژوهش عمده‌تاً بر قابلیت‌های پیاده‌سازی شده و گردش کارهای ارائه‌شده در سیستم استوار بوده و ادعایی مبنی بر بهبود اندازه‌گیری شده در زمینه حفظ میراث فرهنگی، دسترسی به معلومات یا میزان مشارکت کاربران مطرح نمی‌شود.

کلمات کلیدی: میراث فرهنگی؛ مشارکت جمعی؛ مشارکت جامعه؛ حفظ دیجیتالی؛ نظارت و بررسی محتوا.